Software Requirement Specification

Project Title: SENTINEX

AI-Based Industrial Safety Monitoring System

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1. Introduction

This initiative is designed to address a significant challenge faced by industrial factories such as monitoring worker safety through CCTV is slow, tiring, and error prone. Supervisors cannot continuously watch multiple camera feeds, and searching hours of recorded footage after an incident is time-consuming. As a result, violations such as a missing helmet, safety vest, gloves, or goggles often go unnoticed until an accident has already occurred. This document sets the stage by introducing the key objectives and motivations behind SENTINEX, an AI-based industrial safety monitoring system that analyzes live and recorded CCTV footage, detects predefined Personal Protective Equipment (PPE) and safety violations, generates alerts, records incidents, and makes them searchable through a web-based dashboard.

2. Vision Document

2.1. Problem Statement

Table 1: Problem Statement

Category	**Description**
Problem	- Supervisors must continuously watch multiple CCTV feeds, which is difficult, fatiguing, and error prone. - PPE violations (helmet, vest, gloves, goggles, safety shoes) and unsafe activities are missed or noticed too late. - Reviewing a past incident requires manually scrubbing through long, unindexed video recordings.
Affects	- Safety supervisors responsible for monitoring workers. - Factory managers who need accurate incident history. - Workers who are exposed to preventable accidents.
Impact	- Increased risk of workplace injuries and accidents. - Delayed response to unsafe conditions. - Wasted supervisor time on manual footage review. - Weak evidence and records for safety audits and compliance.

Category	Description
Solution	A comprehensive AI-powered industrial safety monitoring system that will: - Process live and recorded CCTV footage. - Detect workers and PPE using computer vision (YOLO). - Identify PPE non-compliance and predefined safety violations. - Track workers across consecutive video frames. - Generate real-time alerts for supervisors. - Store incidents with timestamps, camera information, and snapshots/clips. - Provide a searchable web dashboard for reviewing incidents.

2.2. Business Opportunities

SENTINEX addresses a critical need in industrial safety and presents substantial opportunities. These include deployment in factories, construction sites, warehouses, and manufacturing plants that already own CCTV infrastructure, so that no new camera hardware is needed. Further opportunities include subscription-based licensing per camera or per site, premium features such as advanced analytics and compliance reporting, integration with existing safety management and audit processes, and partnerships with safety consultancies and insurance providers who benefit from reliable incident records.

2.3. Objectives

Following are the objectives of SENTINEX:

- Develop a system to monitor workers and verify the use of required Personal Protective Equipment (PPE).
- Identify predefined workplace safety and PPE violations using AI-based detection.
- Monitor both live and recorded CCTV footage.
- Generate timely alerts for detected safety violations.
- Record important safety incidents with timestamps, camera information, and visual evidence.
- Provide a searchable web-based platform so supervisors can quickly find and review incidents instead of manually searching long recordings.
- Provide a dashboard summary and incident history so managers can review workplace safety status and make informed decisions.

2.4. Scope

The scope of SENTINEX is to provide an AI-based platform that assists supervisors in maintaining workplace safety by automatically analyzing CCTV footage. It connects Safety Supervisors, Factory Managers, and System Admins in a unified monitoring ecosystem.

The core of the system is an AI engine that detects workers and PPE (helmets, safety vests, gloves, goggles, and safety shoes) in live or recorded footage, tracks workers across frames, and checks compliance against predefined rules. Detected violations generate alerts and are stored as incidents with their timestamp, camera information, and a clip.

A web dashboard lets Authorized Users (Factory Managers and Safety Supervisors) view the dashboard, search safety violations, and view incident clips. Safety Supervisors also receive safety alerts and analyse recorded video, while System Admins manage cameras, PPE rules, and users. SENTINEX is a decision-support tool and is not intended to replace human safety supervisors.

2.5. Constraints

Following are the constraints of SENTINEX:

- **Technical:** Detection accuracy may be affected by camera angle, resolution, lighting conditions, occlusion, distance of workers from the camera, and crowded scenes. Real-time processing requires adequate computing power, and detection latency must be kept low for timely alerts.
- **Dataset:** The AI models depend on the quality and variety of available training data (e.g., SH17 Industrial PPE Dataset and other public datasets). Detection of rare or unusual PPE types may be limited.
- **Environmental:** A stable network connection between cameras and the processing server is needed for live monitoring. Poor connectivity may cause dropped frames or delayed alerts.
- **Hardware:** Development and testing are limited to a standard development PC (Windows 10/11, 8 GB RAM minimum, 16 GB recommended) and a webcam or IP camera. Processing many simultaneous camera streams may require a GPU.
- **Privacy and Legal:** Video footage of workers is sensitive. Access to footage and incident records must be restricted to authorized users.
- **Scope of Detection:** The system detects only a predefined set of PPE and safety violations and does not claim to detect every possible hazard.

2.6. Stakeholder and User Description

2.6.1. Market Demographics

The target market comprises industrial organizations that use CCTV for workplace monitoring, including manufacturing factories, construction sites, warehouses, and industrial plants. Primary users are safety supervisors, factory managers, and system administrators working in these environments.

2.6.2. Stakeholder Summary

Table 2: Stakeholder Summary

Name	**Description**	**Responsibility**
Development Team	Fatima Nasir, Meerab Awais, Mufaza Urfan	Fully involved in the analysis, design, development, and testing of the project.
Supervisor	Dr. Ammar Rafiq guides the development team.	Provide technical help, review progress, and support the development team.
Safety Supervisor	The primary user who monitors workers and responds to violations.	Receive and respond to safety alerts, analyse recorded video, search and review incidents.
Factory Manager	Management user responsible for overall workplace safety.	View the dashboard, search and review incident history and footage, and make informed decisions.
System Admin	Manages the system's users, cameras, and configuration.	Manage users and configure cameras and PPE rules.
CCTV Cameras (External System)	Cameras providing live streams and recorded footage.	Supply video input to the system.

2.6.3. User Environment

Users will access SENTINEX through a web-based dashboard in a standard web browser on a desktop or laptop computer, typically from a security or supervision room within the facility. CCTV cameras (IP cameras or webcams during testing) are installed at industrial work areas and connected to the processing server over a local network. A stable network connection is required

for live video processing and alert delivery. Adequate lighting and clear camera views of work areas are ideal for accurate detection.

2.6.4. Stakeholder Profile

2.6.4.1. Supervisor Team

Table 3: Supervisor Team

Representatives	Dr. Ammar Rafiq
Description	Involved in supervising all activities of the development process.
Type	Technical stakeholder with expertise in the project domain.
Responsibility	1. Give direction to the development team. 2. Ensure the system is completed within the given timeline. 3. Keep track of development tasks and progress.
Success Criteria	The successful completion and delivery of the features committed by the development team.

2.6.4.2. Development Team

Table 4: Development Team

Representatives	Fatima Nasir, Meerab Awais, Mufaza Urfan
Description	Involved in the complete design, development, and testing of the project.
Type	Technical stakeholders.
Responsibility	1. Fully involved in the development of the project. 2. Specify and document requirements. 3. Implement system features. 4. Ensure successful development of the proposed system.
Success Criteria	Successful completion, testing, and deployment of all committed features.

2.6.4.3. Safety Supervisor

Table 5: Safety Supervisor

Representatives	Safety officers and shift supervisors.
Description	The primary end users who monitor workers through CCTV and respond to safety violations.
Type	External stakeholder (end user).
Responsibility	Receive and acknowledge violation alerts, analyse recorded video, search and review incidents and incident clips.
Success Criteria	Violations are detected and reported promptly, and past incidents can be found quickly without manual footage scrubbing.

2.6.4.4. Factory Manager

Table 6: Factory Manager

Representatives	Factory managers, plant managers, HSE managers.
Description	Management users who oversee overall workplace safety and compliance.
Type	External stakeholder.
Responsibility	View the dashboard, search and review incident history and footage, and make informed decisions.
Success Criteria	Reliable incident records and footage that support audits and safety improvements.

2.6.4.5. System Admin

Table 7: System Admin

Representatives	Admin / IT staff.
Description	Manage the system's users, cameras, configuration, and overall health so that it functions correctly.
Type	Internal stakeholder.
Responsibility	Manage users and configure cameras and required PPE rules.
Success Criteria	The system runs reliably with correctly configured cameras and authorized users.

3. System Requirement Specification

This section specifies the requirements of SENTINEX. It presents the system's main features, followed by detailed functional and non-functional requirements to ensure that the system fulfills its intended purpose.

3.1. System Features

SENTINEX offers a range of features for its users, including:

- **Provide CCTV Footage:** Receiving live video streams (frames) from configured CCTV cameras for continuous workplace safety monitoring.
- **Analyse CCTV Footage:** YOLO-based detection of workers and PPE (helmets, safety vests, gloves, goggles, and safety shoes) in the received footage.
- **View Dashboard:** A web dashboard showing recent violations and active alerts to authorized users.
- **Search Safety Violations:** Searching violations by date range, camera, and violation type to quickly find relevant footage.
- **View Incident Clips:** Viewing incident details and playing the saved clip or snapshot as evidence.
- **Receive Safety Alerts:** Real-time alerts for Safety Supervisors whenever a predefined violation is detected, with acknowledgement.
- **Analyse Recorded Video:** Analysis of previously recorded CCTV footage by Safety Supervisors to identify violations and important events.
- **Configure Cameras and Rules:** Adding cameras and defining the required PPE rules for each camera or work area.
- **Manage Users:** Creating and managing user accounts and roles, with secure login and role-based access.
- **Track Workers:** Tracking workers across consecutive video frames to maintain continuous monitoring and avoid duplicate alerts.
- **Detect PPE Compliance Violations:** Checking each tracked worker against the required PPE rules to identify violations.
- **Generate Safety Alerts:** Automatic creation of an alert, with its alert ID, violation type, camera, and timestamp, when a violation is detected in live footage.
- **Log Incident:** Stores each incident with its timestamp, violation type and snapshot.

3.2. Functional Requirements

The system shall fulfil specific functional requirements, including:

3.2.1. Provide CCTV Footage

- The system shall receive live video streams (frames) from configured CCTV cameras.
- The system shall accept footage only from cameras that are configured and active.
- The system shall pass the received frames to Analyse CCTV Footage for processing.
- The system shall return the frame processing result to the camera source and continue with the next frames.
- The system shall mark a camera as disconnected when its stream is lost and resume receiving footage when the stream is restored.

3.2.2. Analyse CCTV Footage

- The system shall retrieve the required PPE rules of a camera before processing its frames.
- The system shall continuously analyse the received frames to monitor workplace activities for safety-related events.
- The system shall detect workers in video frames.
- The system shall detect PPE items including helmets, safety vests, gloves, goggles, and safety shoes.
- The system shall associate each detected PPE item with the corresponding worker.
- The system shall pass the detected workers to Track Workers.
- The system shall skip a frame and log the error when detection fails for that frame.

3.2.3. View Dashboard

- The system shall provide a web-based dashboard for authorized users (Factory Managers and Safety Supervisors).
- The system shall retrieve summary data and active alerts from the incident records when the dashboard is opened.
- The system shall display a summary of recent violations and active alerts on the dashboard.
- The system shall allow users to navigate from the dashboard to safety violation search and incident clips.

3.2.4. Search Safety Violations

- The system shall allow authorized users to search safety violations by date range, camera, and violation type.
- The system shall query the incident records using the entered filters and return the matching violations.
- The system shall display the matching violations as a list with timestamps and thumbnails.
- The system shall display a “No incidents found” message when no violation matches the filters.
- The system shall allow the user to select a violation from the list to view its incident clip.

3.2.5. View Incident Clips

- The system shall allow authorized users to select an incident and view its details (camera, timestamp, violation type).
- The system shall retrieve the incident record and its saved clip or snapshot from the incident records.
- The system shall allow the user to play the saved incident clip.
- The system shall display the saved snapshot when no clip exists for the incident.
- The system shall display an error message when the incident clip cannot be loaded.

3.2.6. Receive Safety Alerts

- The system shall display alerts to the Safety Supervisor on the dashboard in near real time.
- The system shall show the alert ID, violation type, camera, and timestamp in each alert received.
- The system shall allow the Safety Supervisor to acknowledge an alert.
- The system shall update the alert status in the incident records when an alert is acknowledged.
- The system shall keep an undelivered or unviewed alert pending and show it the next time the dashboard loads.

3.2.7. Analyse Recorded Video

- The system shall allow the Safety Supervisor to upload a recorded CCTV video file and select its camera.
- The system shall validate the uploaded video file before analysis.
- The system shall analyse the video frames using worker and PPE detection, worker tracking (Track Workers), and PPE compliance violation detection.
- The system shall store the violations detected in recorded video as incidents in the same incident records as live incidents.
- The system shall display an analysis summary to the Safety Supervisor after analysis.
- The system shall display an error message when the video file is unsupported or corrupted.

3.2.8. Configure Cameras and Rules

- The system shall allow the System Admin to open a configuration panel showing the current cameras and PPE rules.
- The system shall allow the System Admin to add or edit a camera (name, stream URL, and location).
- The system shall validate camera details before storing them.
- The system shall allow the System Admin to define the required PPE for each camera or work area.
- The system shall store the camera details and PPE rules and display a confirmation.

3.2.9. Manage Users

- The system shall allow the System Admin to create, edit, and deactivate user accounts.
- The system shall allow the System Admin to assign a role (Factory Manager, Safety Supervisor, or System Admin) to each user.
- The system shall validate user data before saving it.
- The system shall authenticate users by validating their credentials before granting access.
- The system shall restrict access to features based on the authenticated user's role.

3.2.10. Track Workers

- The system shall track detected workers across consecutive video frames.
- The system shall assign a consistent tracking identifier to each worker while the worker remains in view.
- The system shall close the track of a worker who leaves the camera view.
- The system shall pass the tracked workers to Detect PPE Compliance Violations.

3.2.11. Detect PPE Compliance Violations

- The system shall compare the detected PPE of each tracked worker with the required PPE rules of the camera.
- The system shall identify predefined PPE violations (e.g., no helmet, no vest, no gloves, no goggles).
- The system shall log an incident for every detected violation (Log Incident).
- The system shall generate a safety alert for violations detected in live footage (Generate Safety Alerts).
- The system shall skip the compliance check and log the error when the PPE rules of a camera are unavailable.

3.2.12. Generate Safety Alerts

- The system shall generate an alert whenever a predefined safety violation is detected in live footage.
- The system shall include the alert ID, violation type, camera, and timestamp in each alert.
- The system shall send each generated alert to the dashboard of the Safety Supervisor (Receive Safety Alerts).
- The system shall not generate repeated alerts for the same continuing violation of the same tracked worker.
- The system shall keep the alert stored with its incident when delivery to the dashboard fails.

3.2.13. Log Incident

- The system shall record each incident with its camera ID, timestamp, and violation type.
- The system shall save a snapshot and/or short video clip as evidence for each incident.
- The system shall return an incident ID for each logged incident.
- The system shall store incidents from live footage and recorded video in the same incident records for later search and review.

3.3. Non-Functional Requirements

This section defines the essential non-functional requirements of SENTINEX:

3.3.1. User Friendly Interface

- The dashboard must feature a clean, intuitive, and clear interface that supervisors can use with minimal training.
- Alerts must be clearly visible and distinguishable from routine information.

3.3.2. Data Security

- The system must protect user data and CCTV footage using secure authentication, hashed passwords, and role-based access control.
- All communication between the dashboard, backend, and database must be encrypted (e.g., HTTPS).
- Only authorized users shall be able to view incident footage and records.

3.3.3. Performance

- The system shall process live video at a rate sufficient for timely detection and shall generate alerts within a few seconds of a violation.
- Searching incidents shall return results promptly, even with a large number of stored incidents.
- The dashboard shall remain responsive while video processing is running.

3.3.4. Accuracy

- The detection models shall achieve acceptable accuracy for workers and PPE under typical industrial conditions, and the system shall aim to minimize false alerts and missed violations.

3.3.5. Scalability

- The system shall be designed so that additional cameras can be added without redesigning the system, subject to available computing resources.

3.3.6. Maintainability

- The system shall use a modular architecture (detection, tracking, backend, database, dashboard) so that components can be updated or improved independently.
- Updates or bug fixes shall be deployable with minimal downtime.

3.3.7. Reliability

- The system shall execute all key operations, including detection, alert generation, incident logging, and incident search, reliably and consistently.
- The system shall handle temporary camera disconnections gracefully and resume processing when the stream is restored.

3.3.8. Compatibility

- The web dashboard shall work on modern web browsers.
- The system shall support common video sources, including IP camera streams, webcams, and standard recorded video file formats.

4. Use Cases

4.1. Use Case Diagram

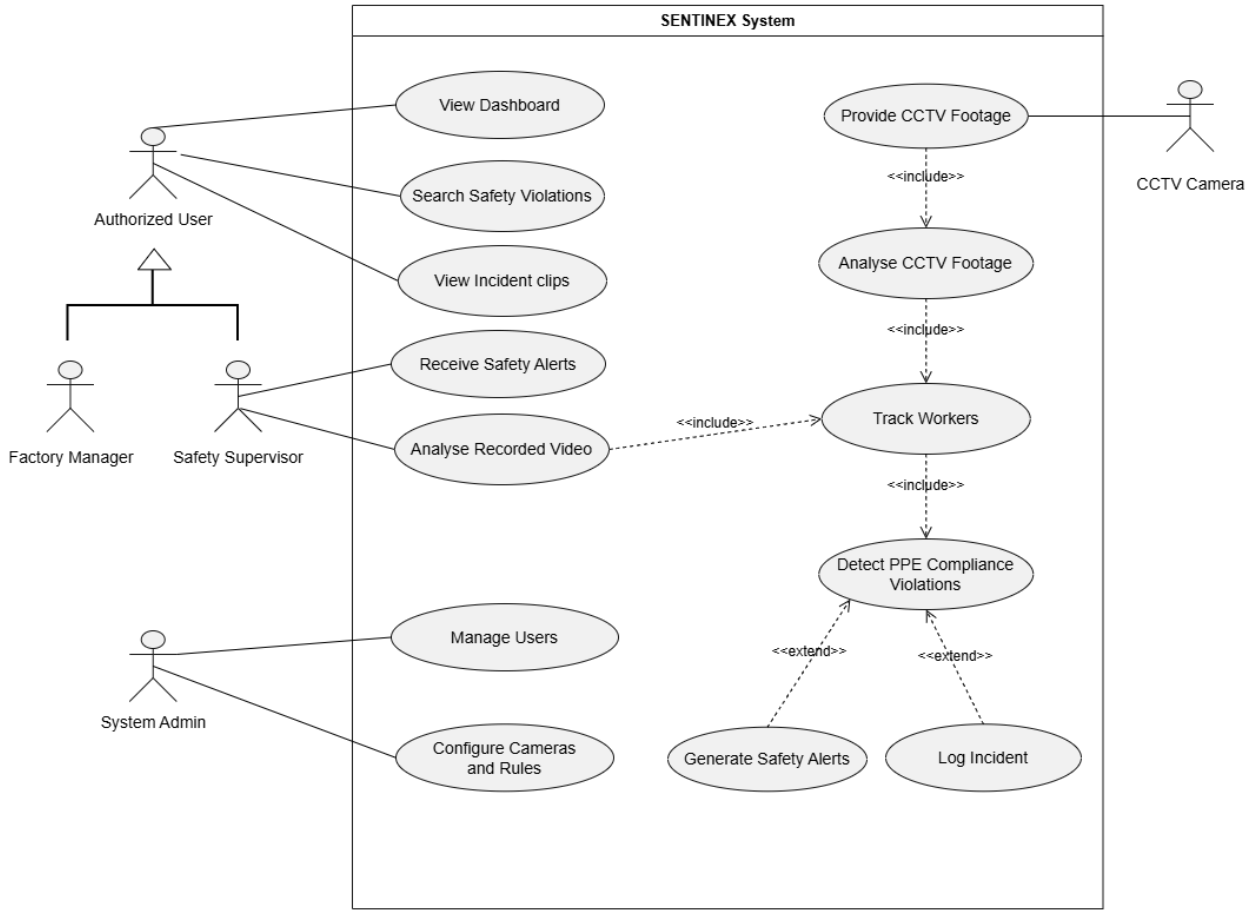


Figure 1: Use Case Diagram

4.2. Expanded Use Cases

4.2.1. Provide CCTV Footage

Table 8: UC-01 Provide CCTV Footage

Use Case Id	UC-01
Use Case	Provide CCTV Footage
Actors	CCTV Camera
Description / Goal	This use case allows a configured CCTV camera to supply live video footage (frames) to the system for continuous safety monitoring.
Trigger / Initiating Event	A configured CCTV camera starts sending its video stream (frames) to the system.
Preconditions	1. The camera is configured and active. 2. The network connection between the camera and the system is available.
Postconditions	1. The video frames are received and passed on for analysis. 2. The camera status (connected or disconnected) is up to date.
Main Flow	1. The camera sends the video stream to the system. 2. The system checks that the camera is configured and active. 3. The system receives the frames. 4. The system passes the frames to Analyse CCTV Footage. 5. The system returns the frame result to the camera source and continues with the next frames.
Alternative Flows	1. Inactive Camera: Footage from an unconfigured or inactive camera is rejected and not processed.
Exception Flow	1. Stream Lost: The system marks the camera as disconnected and resumes receiving footage when the stream is restored.
Includes	Analyse CCTV Footage
Extends	None
Frequency of Use	Continuous. This is the entry point of the core operation of the system.
Business Rules / Constraints	1. Only active, configured cameras are processed.
Assumptions	Cameras provide a clear view of work areas with adequate lighting and a stable network connection.
Notes / Issues	Analyse CCTV Footage further includes Track Workers, which includes Detect PPE Compliance Violations.

4.2.2. Analyse CCTV Footage

Table 9: UC-02 Analyse CCTV Footage

Use Case Id	UC-02
Use Case	Analyse CCTV Footage
Actors	None (Internal System)
Description / Goal	This use case analyses the received live footage to detect workers and PPE items in the video frames.
Trigger / Initiating Event	Frames are received from Provide CCTV Footage.
Preconditions	1. Frames are available. 2. Required PPE rules are defined for the camera. 3. The detection models are loaded.
Postconditions	1. Workers and PPE items are detected and associated with each other. 2. The detections are passed to Track Workers.
Main Flow	1. The system receives the frames. 2. The system retrieves the required PPE rules for that camera. 3. The system detects workers in each frame. 4. The system detects PPE items (helmet, vest, gloves, goggles, safety shoes). 5. The system associates the PPE items with the workers. 6. The system passes the detections to Track Workers. 7. The system returns the result and continues with the next frames.
Alternative Flows	1. No Worker: If no worker is found in a frame, an empty result is returned and processing continues.
Exception Flow	1. Frame Processing Failure: If detection fails for a frame, the frame is skipped and the error is logged.
Includes	Track Workers
Extends	None
Frequency of Use	Continuous during live monitoring.
Business Rules / Constraints	1. Detections below the confidence threshold are ignored. 2. Only the predefined PPE items are detected.
Assumptions	The detection model is trained on suitable PPE data.
Notes / Issues	Included by Provide CCTV Footage.

4.2.3. View Dashboard

Table 10: UC-03 View Dashboard

Use Case Id	UC-03
Use Case	View Dashboard
Actors	Authorized User (Factory Manager, Safety Supervisor)
Description / Goal	This use case allows an authorized user to see a summary of recent violations and active alerts.
Trigger / Initiating Event	The user logs in and opens the dashboard.
Preconditions	1. The user must be logged in. 2. The user has the Factory Manager or Safety Supervisor role.
Postconditions	1. The dashboard summary and active alerts are displayed.
Main Flow	1. The user opens the dashboard. 2. The system requests the dashboard data. 3. The system fetches the summary and active alerts from the incident records. 4. The system returns the dashboard data. 5. The system displays the dashboard to the user.
Alternative Flows	1. No Data: If no incidents or alerts exist, the dashboard shows an empty summary.
Exception Flow	1. Data Retrieval Error: The system displays “Could not load dashboard. Please try again.”
Includes	None
Extends	None
Frequency of Use	High. Opened at the start of each session.
Business Rules / Constraints	1. Only authorized users can view the dashboard. 2. The dashboard is read-only.
Assumptions	Incidents and alerts have been generated by monitoring.
Notes / Issues	The dashboard is the entry point to Search Safety Violations and View Incident Clips.

4.2.4. Search Safety Violations

Table 11: UC-04 Search Safety Violations

Use Case Id	UC-04
Use Case	Search Safety Violations
Actors	Authorized User (Factory Manager, Safety Supervisor)
Description / Goal	This use case allows an authorized user to quickly find detected safety violations without manually searching complete recordings.
Trigger / Initiating Event	The user opens the safety violation search page from the dashboard.
Preconditions	1. The user must be logged in with an authorized role. 2. Incidents have been logged in the system.
Postconditions	1. A list of matching violations is displayed to the user.
Main Flow	1. The user opens the safety violation search page. 2. The user enters search filters (date range, camera, violation type). 3. The system queries the incident records using the filters. 4. The system returns the matching violations. 5. The system displays the violation list with timestamps and thumbnails. 6. The user may select a violation to view its clip (View Incident Clips).
Alternative Flows	1. No Results: The system displays “No incidents found.” 2. No Filters: The system displays the most recent violations.
Exception Flow	1. Data Retrieval Error: The system displays “Could not load incidents. Please try again.”
Includes	None
Extends	None
Frequency of Use	Medium to High. Used after alerts and during safety reviews.
Business Rules / Constraints	1. Only authorized users can search violations. 2. Search results are read-only.
Assumptions	Incidents have already been logged by live monitoring or recorded video analysis.
Notes / Issues	Selecting a result leads to View Incident Clips.

4.2.5. View Incident Clips

Table 12: UC-05 View Incident Clips

Use Case Id	UC-05
Use Case	View Incident Clips
Actors	Authorized User (Factory Manager, Safety Supervisor)
Description / Goal	This use case allows an authorized user to view the details and saved clip of a selected incident as evidence.
Trigger / Initiating Event	The user selects an incident from the violation list or the dashboard.
Preconditions	1. The user must be logged in. 2. The incident exists with a saved clip or snapshot.
Postconditions	1. The incident details and clip are displayed to the user.
Main Flow	1. The user selects an incident. 2. The system requests the incident clip. 3. The system fetches the incident details and clip from the incident records. 4. The system returns the details and clip. 5. The system displays the incident details (camera, timestamp, violation type). 6. The system plays the incident clip.
Alternative Flows	1. Snapshot Only: If no clip exists, the system displays the saved snapshot.
Exception Flow	1. Clip Load Failure: The system displays “Could not load incident clip.”
Includes	None
Extends	None
Frequency of Use	Medium. After searching or receiving an alert.
Business Rules / Constraints	1. Clips are accessible only to authorized users. 2. Users cannot modify or delete incident evidence.
Assumptions	The clip was saved when the incident was logged.
Notes / Issues	None

4.2.6. Receive Safety Alerts

Table 13: UC-06 Receive Safety Alerts

Use Case Id	UC-06
Use Case	Receive Safety Alerts
Actors	Safety Supervisor
Description / Goal	This use case allows the Safety Supervisor to receive an alert whenever a predefined safety violation is detected and to acknowledge it.
Trigger / Initiating Event	The system sends an alert, created by Generate Safety Alerts, to the dashboard.
Preconditions	1. The supervisor is logged in with the dashboard open. 2. An alert has been generated for a detected violation.
Postconditions	1. The alert is displayed to the supervisor. 2. After acknowledgement, the alert status is updated to acknowledged.
Main Flow	1. The system sends the alert (alert ID, violation type, camera, and timestamp) to the dashboard. 2. The supervisor views the alert. 3. The supervisor acknowledges the alert. 4. The system updates the alert status in the database. 5. The system displays the acknowledgement.
Alternative Flows	1. Supervisor Offline: The alert stays pending until it is viewed.
Exception Flow	1. Alert Delivery Failure: The alert remains stored with the incident and is shown the next time the dashboard loads.
Includes	None
Extends	None
Frequency of Use	Depends on the number of violations; high during active monitoring.
Business Rules / Constraints	1. Alerts are shown only for predefined violations. 2. An alert stays active until it is acknowledged.
Assumptions	Supervisors monitor the dashboard during working hours.
Notes / Issues	The alerts received here are created by Generate Safety Alerts.

4.2.7. Analyse Recorded Video

Table 14: UC-07 Analyse Recorded Video

Use Case Id	UC-07
Use Case	Analyse Recorded Video
Actors	Safety Supervisor
Description / Goal	This use case allows the Safety Supervisor to analyse previously recorded CCTV footage to find safety violations and important events.
Trigger / Initiating Event	The supervisor selects Upload Video and chooses a video file and camera.
Preconditions	1. The supervisor must be logged in. 2. The camera is configured with PPE rules. 3. The video file is in a supported format.
Postconditions	1. Detected violations are stored as incidents. 2. An analysis summary is displayed to the supervisor.
Main Flow	1. The supervisor uploads the video file and selects the camera. 2. The system submits the video for analysis. 3. The system validates the video file. 4. The system detects workers and PPE in the video frames. 5. The system tracks workers across frames (Track Workers). 6. The system detects PPE compliance violations using the camera's rules. 7. The system logs the incidents (Log Incident). 8. The system returns the analysis summary. 9. The system displays the summary to the supervisor.
Alternative Flows	1. No Violations: The summary shows zero incidents.
Exception Flow	1. Invalid File: The system displays "Unsupported or corrupted video file."
Includes	Track Workers
Extends	None
Frequency of Use	Low to Medium. As needed for reviewing past footage.
Business Rules / Constraints	1. Only Safety Supervisors can upload recorded videos. 2. Incidents from recorded video are stored with the live incidents. 3. Safety alerts are generated only for live footage.
Assumptions	The recorded footage has acceptable quality and clear camera views.
Notes / Issues	Analysis time depends on video length and available processing power.

4.2.8. Configure Cameras and Rules

Table 15: UC-08 Configure Cameras and Rules

Use Case Id	UC-08
Use Case	Configure Cameras and Rules
Actors	System Admin
Description / Goal	This use case allows the admin to add or edit cameras and define the required PPE rules used for violation detection.
Trigger / Initiating Event	The admin opens the configuration panel.
Preconditions	1. The user must be logged in with Admin privileges.
Postconditions	1. Camera details and PPE rules are stored. 2. Updated rules are applied to subsequent monitoring.
Main Flow	1. The admin accesses the configuration panel. 2. The system displays the current cameras and rules. 3. The admin enters the camera name, stream URL, and location and saves. 4. The system validates and stores the camera details. 5. The system displays a confirmation. 6. The admin selects a camera and updates its required PPE. 7. The system saves the rules. 8. The system displays a confirmation.
Alternative Flows	1. Invalid Input: If required fields are missing, the system shows a validation error and does not save.
Exception Flow	1. Database Error: The system displays “Configuration could not be saved. Please try again.”
Includes	None
Extends	None
Frequency of Use	Low. When cameras or rules change.
Business Rules / Constraints	1. Only users with the Admin role can access this function. 2. PPE rules are defined per camera or work area.
Assumptions	The admin knows the camera stream address and the PPE required in each area.
Notes / Issues	None

4.2.9. Manage Users

Table 16: UC-09 Manage Users

Use Case Id	UC-09
Use Case	Manage Users
Actors	System Admin
Description / Goal	This use case allows the admin to create, edit, or deactivate user accounts and assign roles.
Trigger / Initiating Event	The admin opens the user management section.
Preconditions	1. The user must be logged in with Admin privileges.
Postconditions	1. The user account change is stored.
Main Flow	1. The admin selects a user action (add, edit, or deactivate) and enters the user details. 2. The system validates the user data. 3. The system saves the user change. 4. The system returns the result. 5. The system displays a confirmation.
Alternative Flows	1. Duplicate User: If the email already exists, the system rejects the change and shows a message.
Exception Flow	1. Database Error: The system displays “User could not be saved. Please try again.”
Includes	None
Extends	None
Frequency of Use	Low. When staff change.
Business Rules / Constraints	1. Only the Admin role can manage users. 2. Each user has exactly one role.
Assumptions	The admin has the details of the users to be added.
Notes / Issues	None

4.2.10. Track Workers

Table 17: UC-10 Track Workers

Use Case Id	UC-10
Use Case	Track Workers
Actors	None (Internal System)
Description / Goal	This use case follows each detected worker across consecutive frames.
Trigger / Initiating Event	Detections are available from Analyse CCTV Footage or Analyse Recorded Video.
Preconditions	1. Detections of workers are available.
Postconditions	1. Each worker has a consistent tracking identifier. 2. The tracked workers are passed to Detect PPE Compliance Violations.
Main Flow	1. The system receives the detections. 2. The system matches the detections with workers from earlier frames. 3. The system assigns or updates tracking identifiers. 4. The system passes the tracked workers to Detect PPE Compliance Violations.
Alternative Flows	1. New Worker: A new tracking identifier is assigned.
Exception Flow	1. Lost Track: If a worker leaves the view, the track is closed.
Includes	Detect PPE Compliance Violations
Extends	None
Frequency of Use	Continuous during monitoring.
Business Rules / Constraints	1. Tracking identifiers are unique within a stream.
Assumptions	Workers remain visible across consecutive frames.
Notes / Issues	Included by Analyse CCTV Footage and Analyse Recorded Video. Tracking supports duplicate alert suppression.

4.2.11. Detect PPE Compliance Violations

Table 18: UC-11 Detect PPE Compliance Violations

Use Case Id	UC-11
Use Case	Detect PPE Compliance Violations
Actors	None (Internal System)
Description / Goal	This use case checks each tracked worker against the required PPE rules and identifies violations.
Trigger / Initiating Event	Tracked workers are available from Track Workers.
Preconditions	1. Tracked workers are available. 2. The required PPE rules of the camera are known.
Postconditions	1. Violations are identified and passed to Log Incident and, for live footage, Generate Safety Alerts.
Main Flow	1. The system receives the tracked workers and the required PPE. 2. The system compares each worker's detected PPE with the rules. 3. The system identifies missing PPE as violations. 4. The system logs the violation as an incident (Log Incident). 5. For live footage, the system generates an alert (Generate Safety Alerts).
Alternative Flows	1. Compliant Worker: No violation is created.
Exception Flow	1. Rules Unavailable: The system skips the check and logs the error.
Includes	None
Extends	None
Frequency of Use	Continuous during monitoring.
Business Rules / Constraints	1. Only predefined violations are detected.
Assumptions	The PPE rules are configured for each camera.
Notes / Issues	Extended by Log Incident and Generate Safety Alerts when a violation is detected. Included by Track Workers.

4.2.12. Generate Safety Alerts

Table 19: UC-12 Generate Safety Alerts

Use Case Id	UC-12
Use Case	Generate Safety Alerts
Actors	None (Internal System)
Description / Goal	This use case creates an alert for the Safety Supervisor whenever a predefined violation is detected in live footage.
Trigger / Initiating Event	A PPE compliance violation is detected in live footage (Detect PPE Compliance Violations).
Preconditions	1. Live monitoring is active. 2. A violation has been identified for a tracked worker.
Postconditions	1. An alert is created and sent to the dashboard of the Safety Supervisor.
Main Flow	1. The system receives the detected violation. 2. The system checks that the violation is not a repeat for the same tracked worker. 3. The system creates an alert with alert ID, violation type, camera, and timestamp. 4. The system sends the alert to the dashboard (Receive Safety Alerts).
Alternative Flows	1. Repeated Violation: If the same tracked worker continues the same violation, no duplicate alert is generated. 2. Recorded Source: Violations found in recorded video do not generate alerts.
Exception Flow	1. Alert Delivery Failure: The alert remains stored with the incident and is shown the next time the dashboard loads.
Includes	None
Extends	Detect PPE Compliance Violations
Frequency of Use	Depends on the number of violations; high during active monitoring.
Business Rules / Constraints	1. Alerts are generated only for predefined violations. 2. Repeated alerts for the same continuing violation are suppressed.
Assumptions	Tracking identifiers are consistent across frames.
Notes / Issues	Extends Detect PPE Compliance Violations when a violation is found in live footage.

4.2.13. Log Incident

Table 20: UC-13 Log Incident

Use Case Id	UC-13
Use Case	Log Incident
Actors	None (Internal System)
Description / Goal	This use case records a detected violation with evidence for later search and review.
Trigger / Initiating Event	A violation is detected (Detect PPE Compliance Violations).
Preconditions	1. A violation has been identified.
Postconditions	1. The incident record with its clip or snapshot is stored.
Main Flow	1. The system receives the violation data. 2. The system saves the snapshot or clip. 3. The system stores the incident with camera ID, timestamp, and violation type. 4. The system returns the incident ID.
Alternative Flows	1. Recorded Source: Incidents from recorded video use the video timestamp.
Exception Flow	1. Storage Failure: The system retries and logs the error.
Includes	None
Extends	Detect PPE Compliance Violations
Frequency of Use	Each time a violation is detected.
Business Rules / Constraints	1. Incident records cannot be edited by users.
Assumptions	Sufficient storage is available.
Notes / Issues	Extends Detect PPE Compliance Violations when a violation is found.

5. Activity Diagram

5.1. Monitoring and Violation Detection

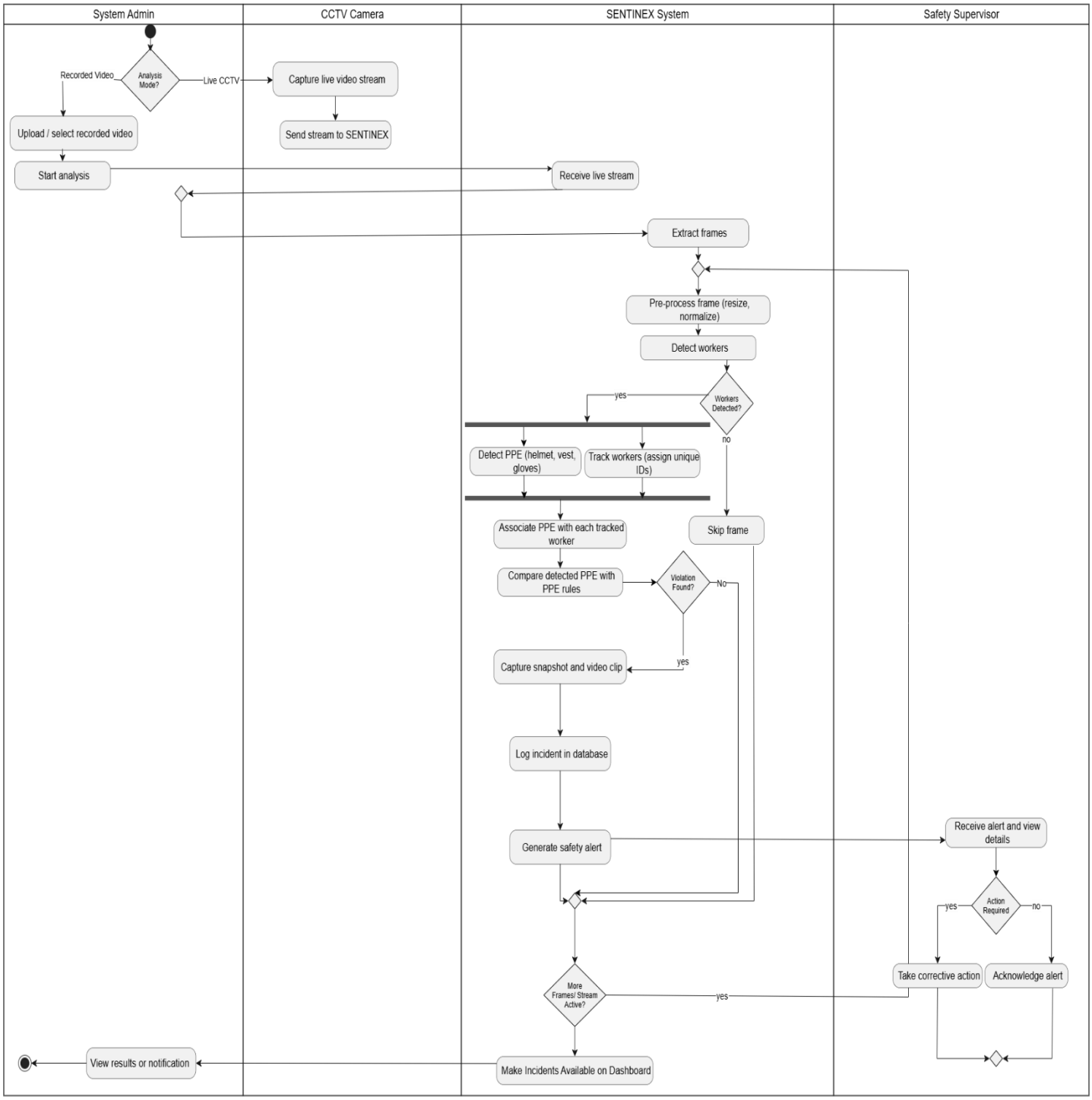


Figure 2: Activity Diagram - Monitoring and Violation Detection

5.2. Incident Review

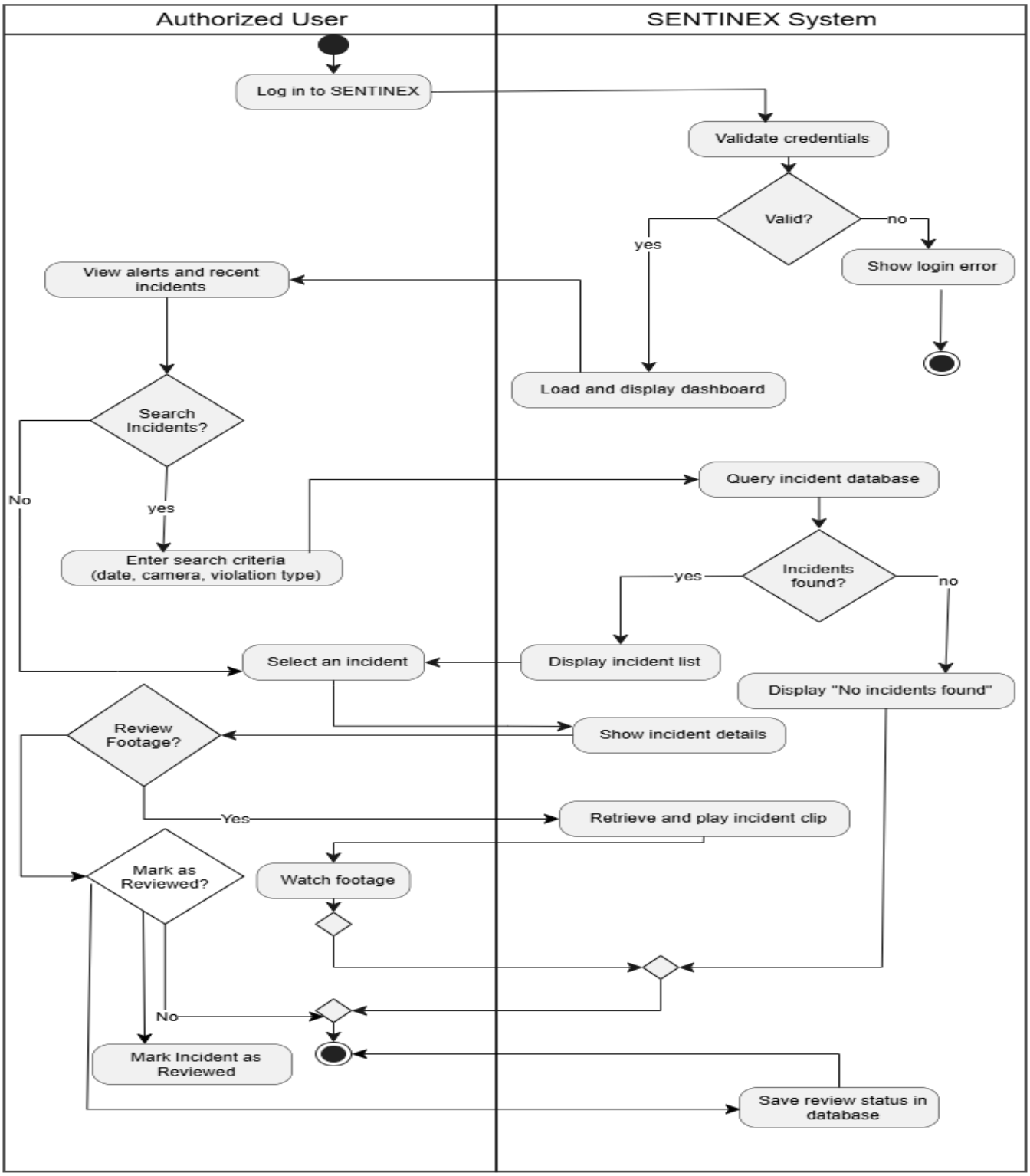


Figure 3: Activity Diagram - Incident Review

5.3. System Administration

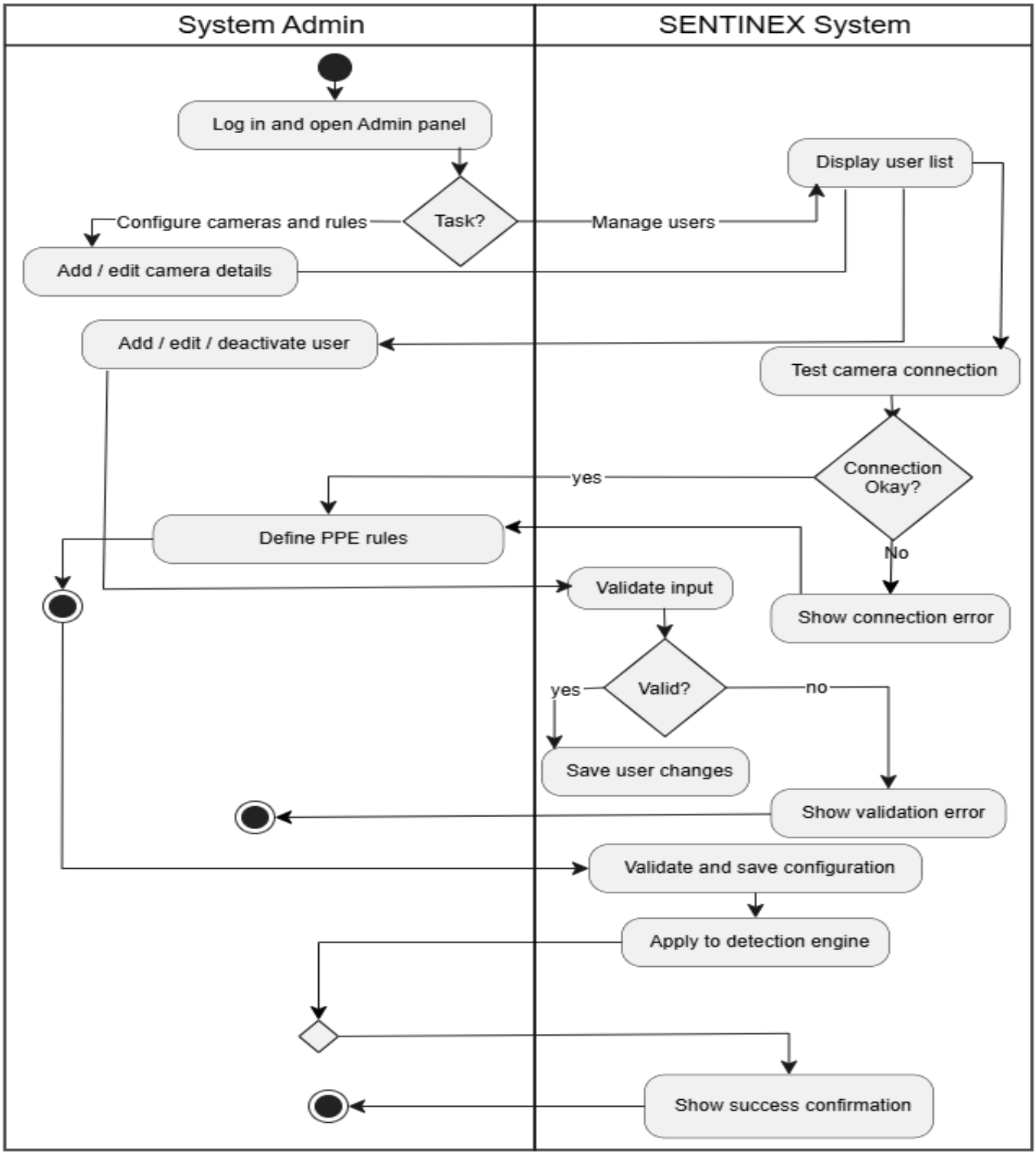


Figure 4: Activity Diagram - System Administration

6. System Sequence Diagram

6.1. Analyse CCTV Footage

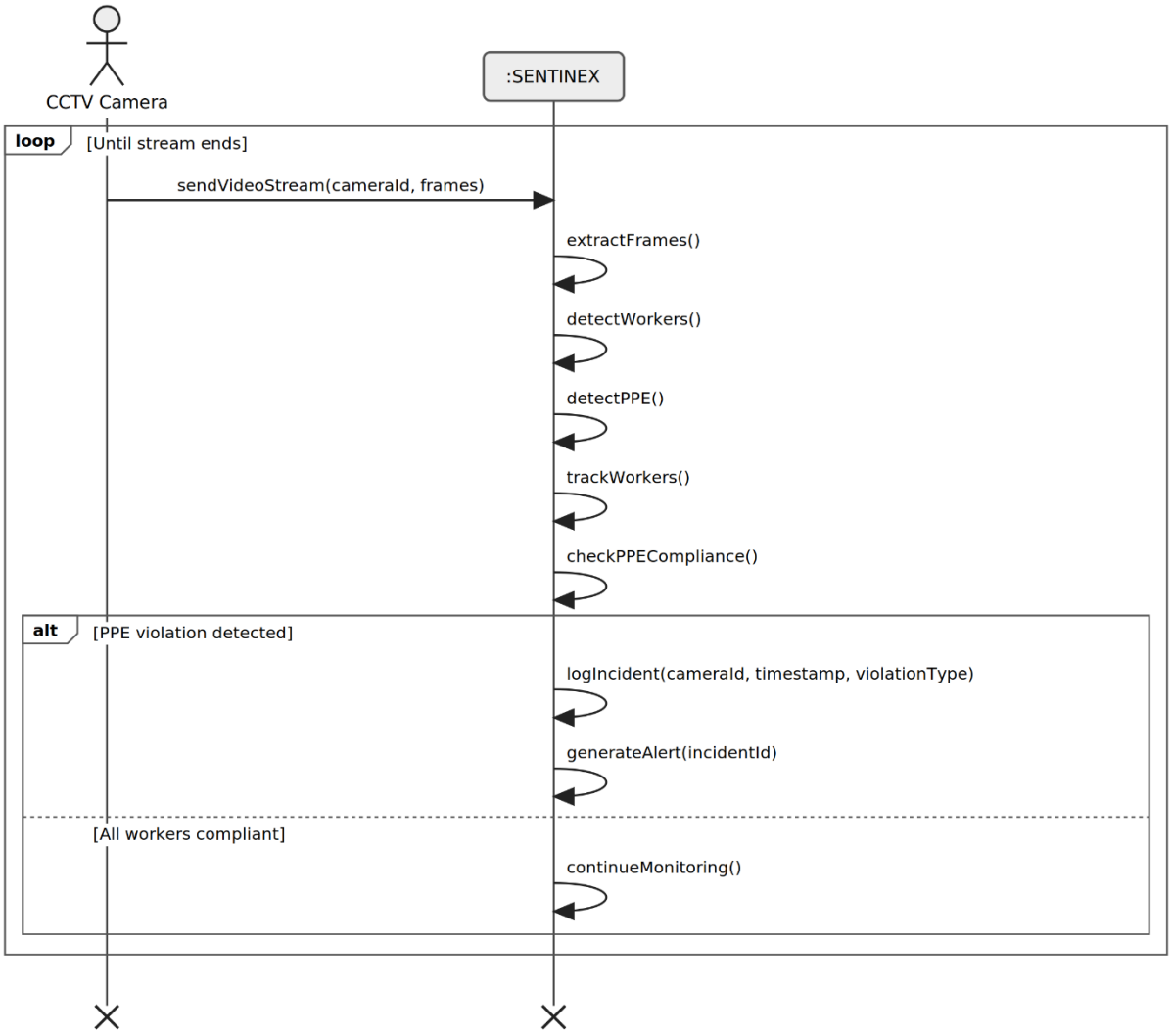


Figure 5: SSD for UC-02 Analyse CCTV Footage

6.2. View Dashboard

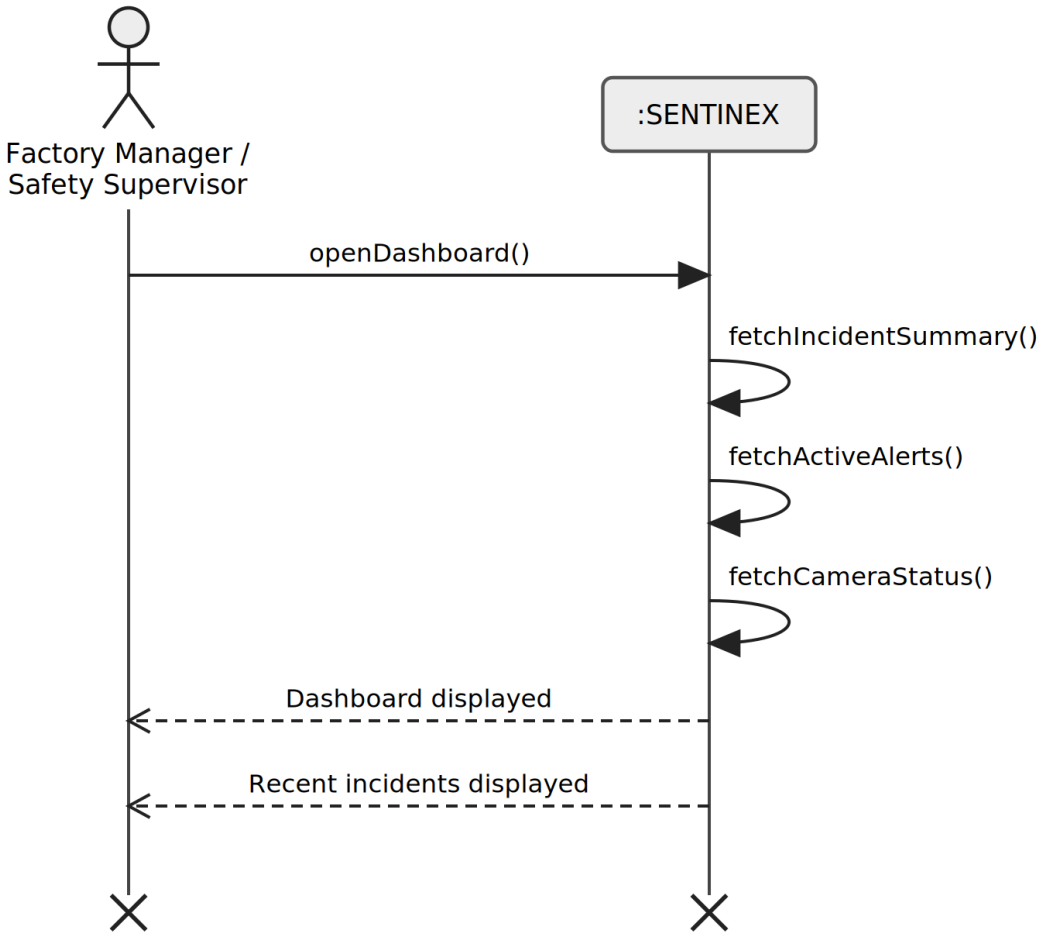


Figure 6: SSD for UC-03 View Dashboard

6.3. Search Safety Violations

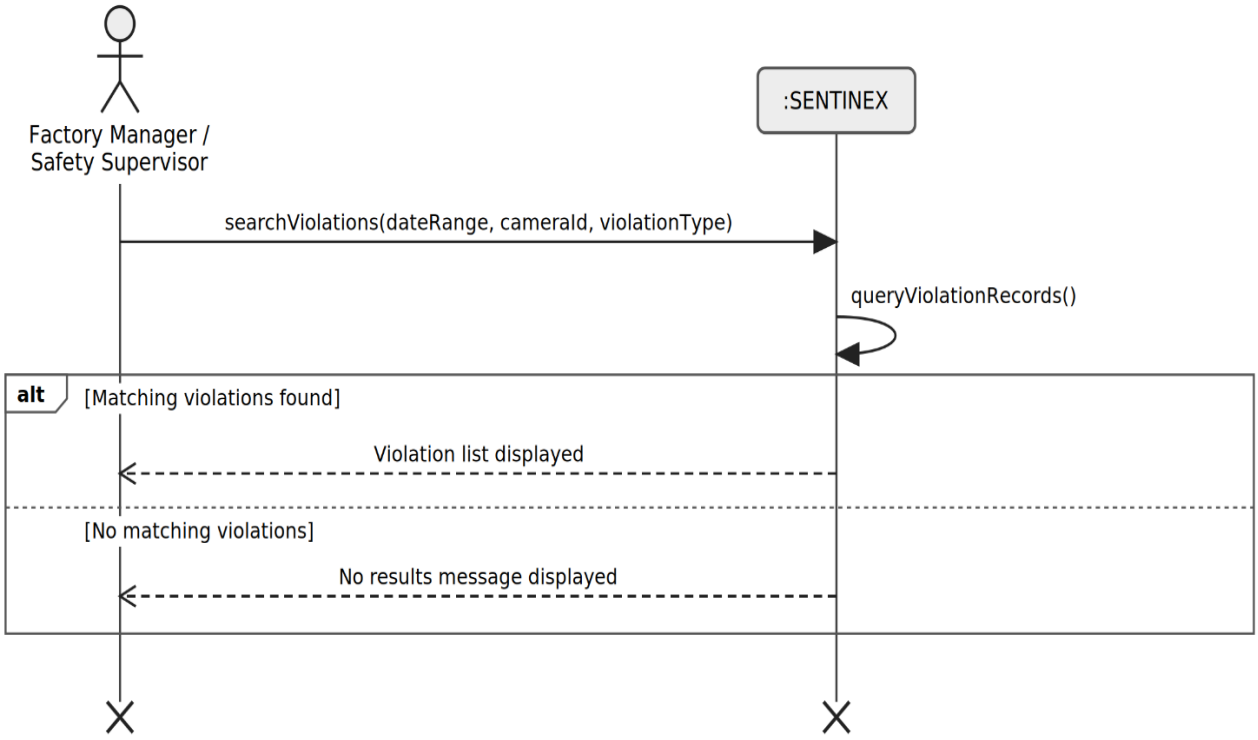


Figure 7: SSD for UC-04 Search Safety Violations

6.4. View Incident Clips

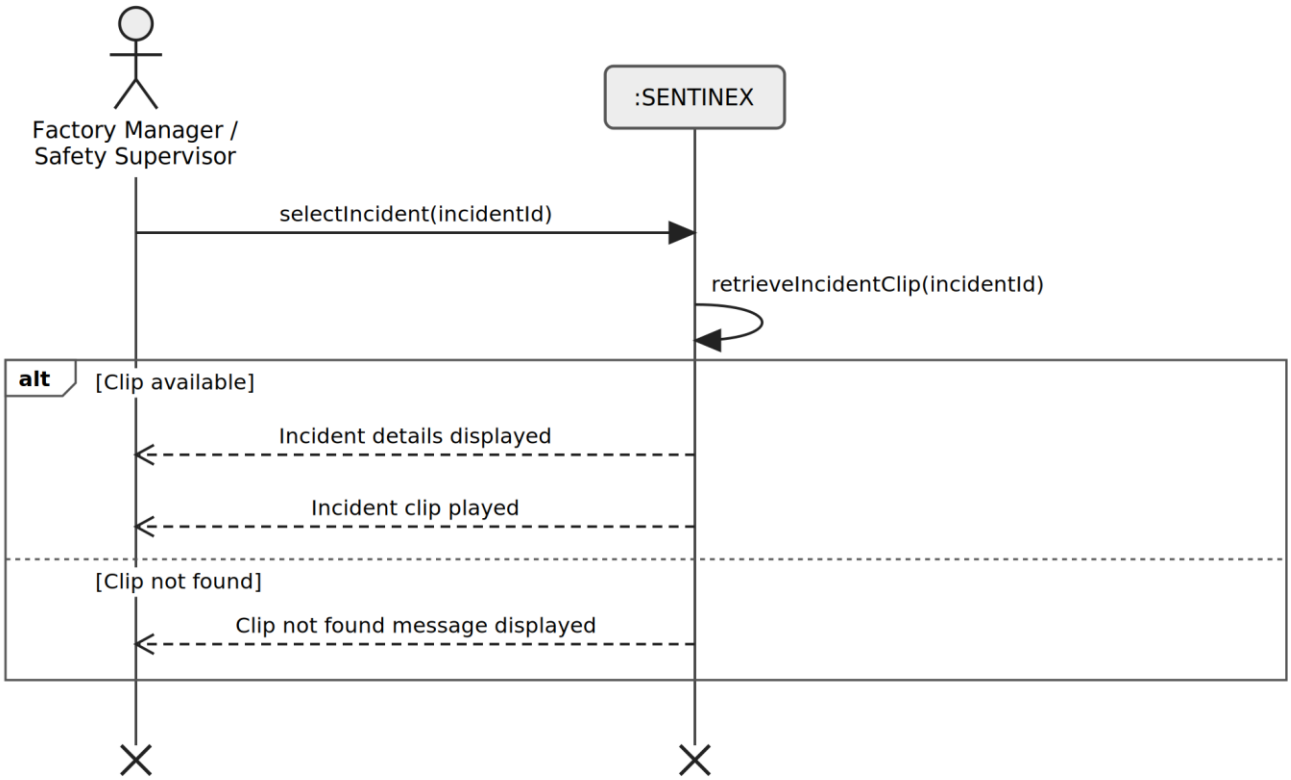


Figure 8: SSD for UC-05 View Incident Clips

6.5. Analyse Recorded Video

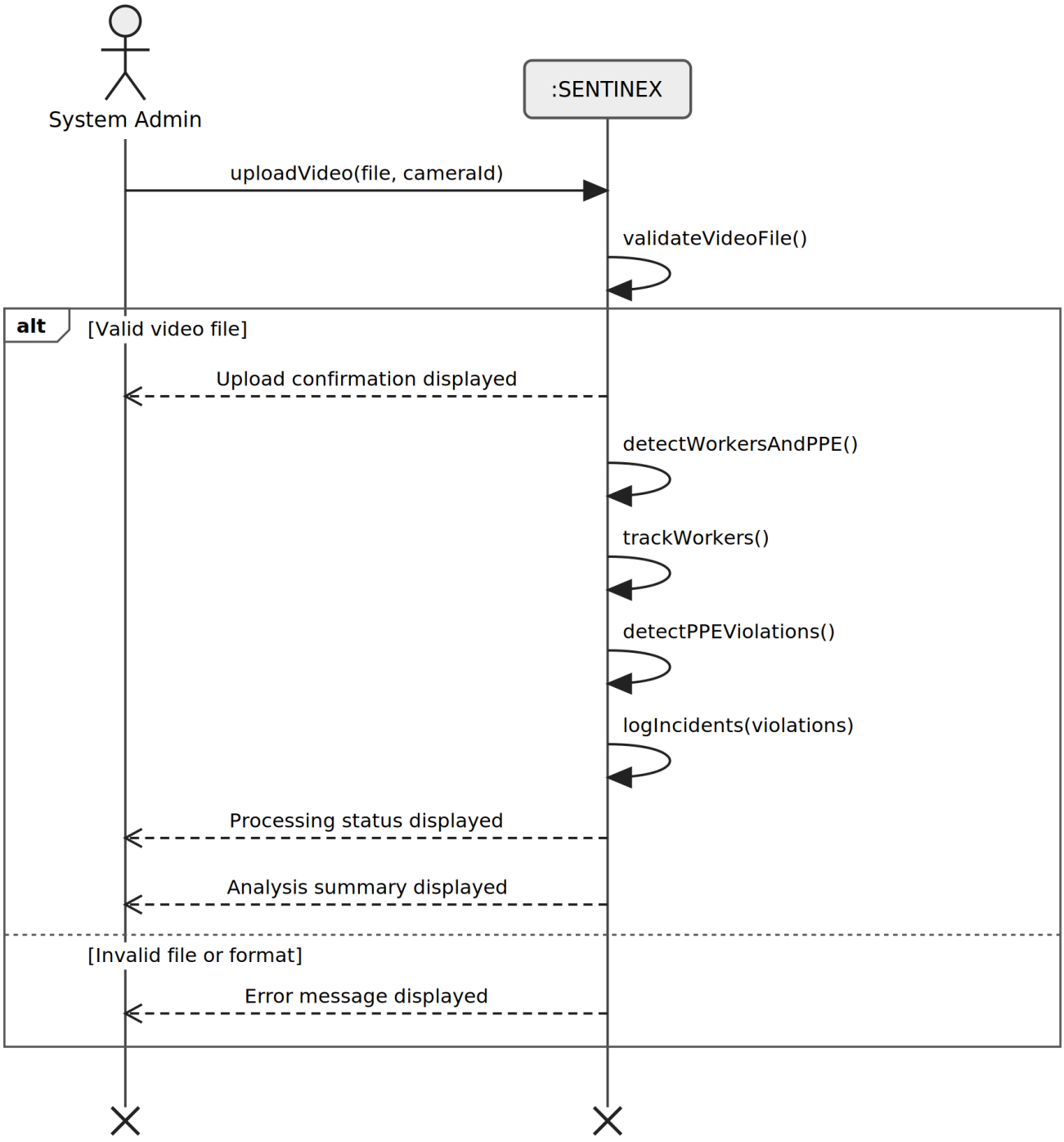


Figure 9: SSD for UC-07 Analyse Recorded Video

6.6. Configure Camera and Rules

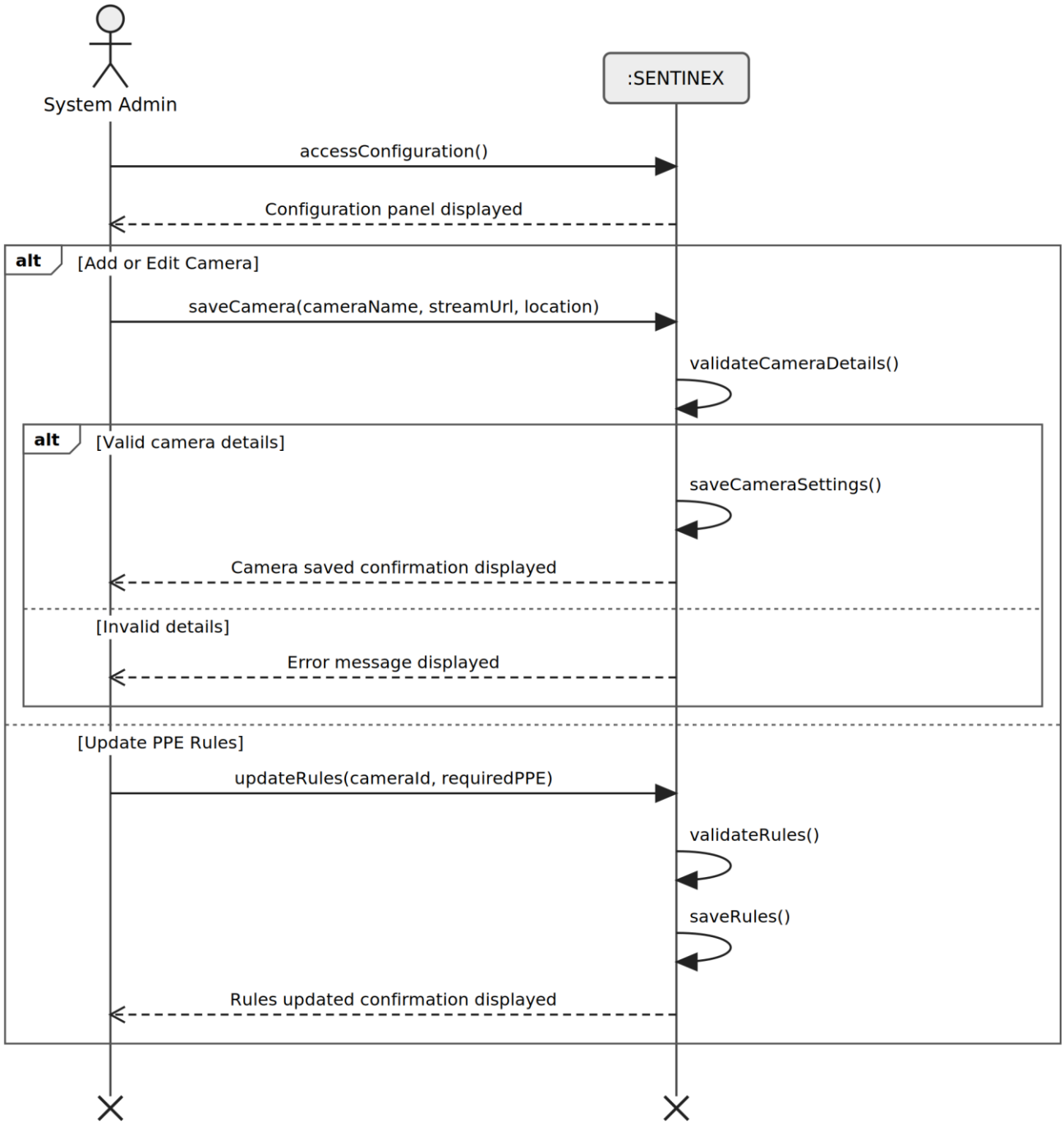


Figure 10: SSD for UC-08 Configure Camera and Rules

6.7. Manage Users

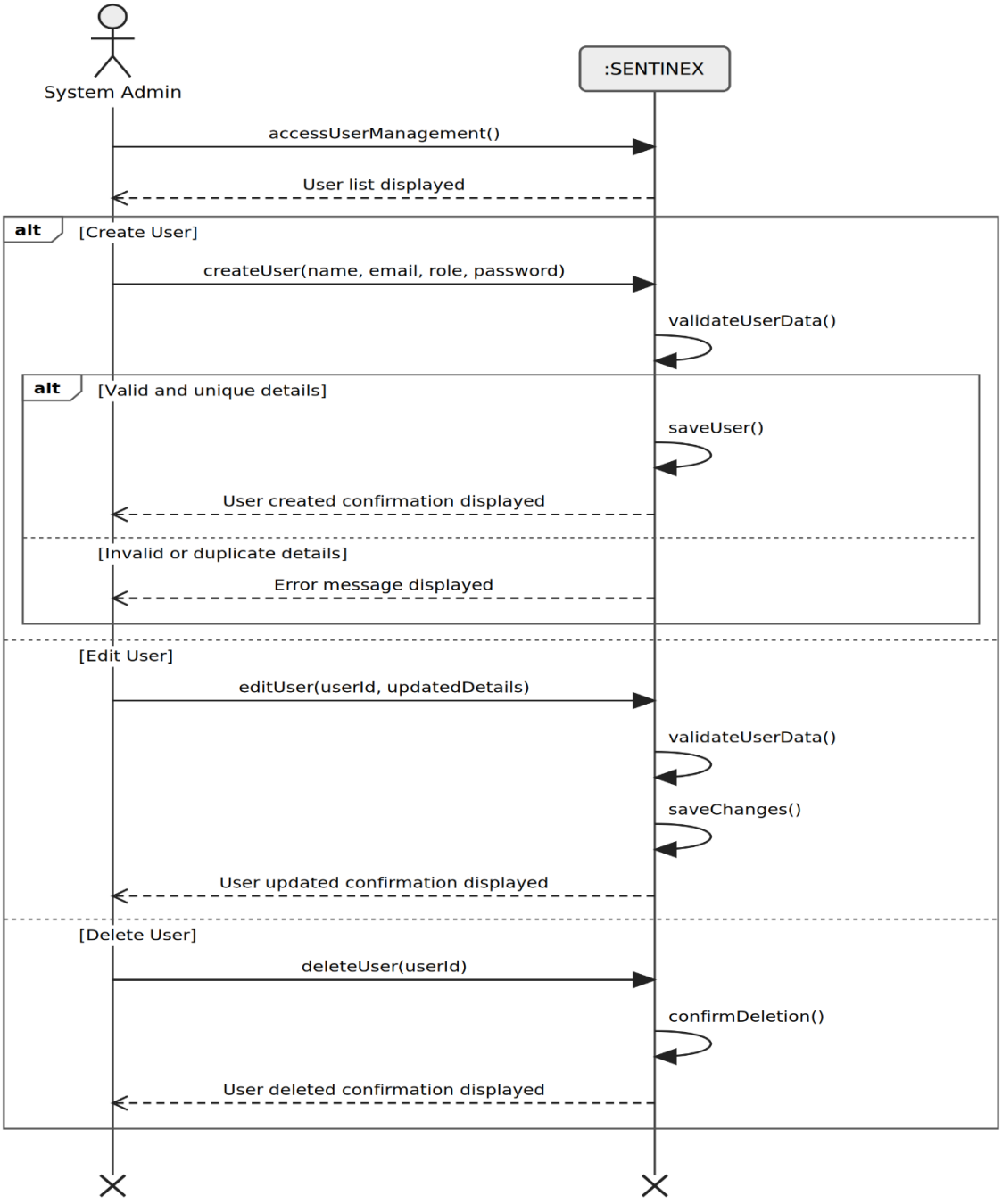


Figure 11: SSD for UC-09 Manage Users

6.8. Generate Safety Alerts

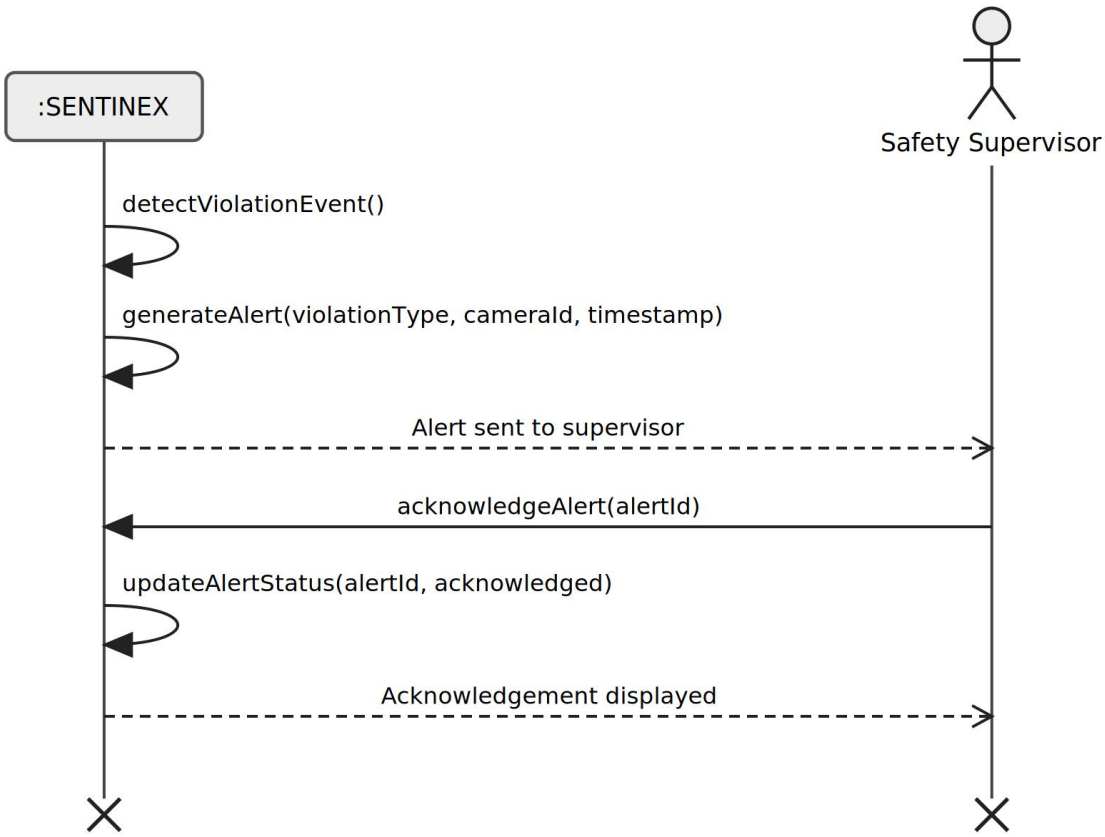


Figure 12: SSD for UC-12 Generate Safety Alerts

7. Structure Design

7.1. High Level Architecture Diagram

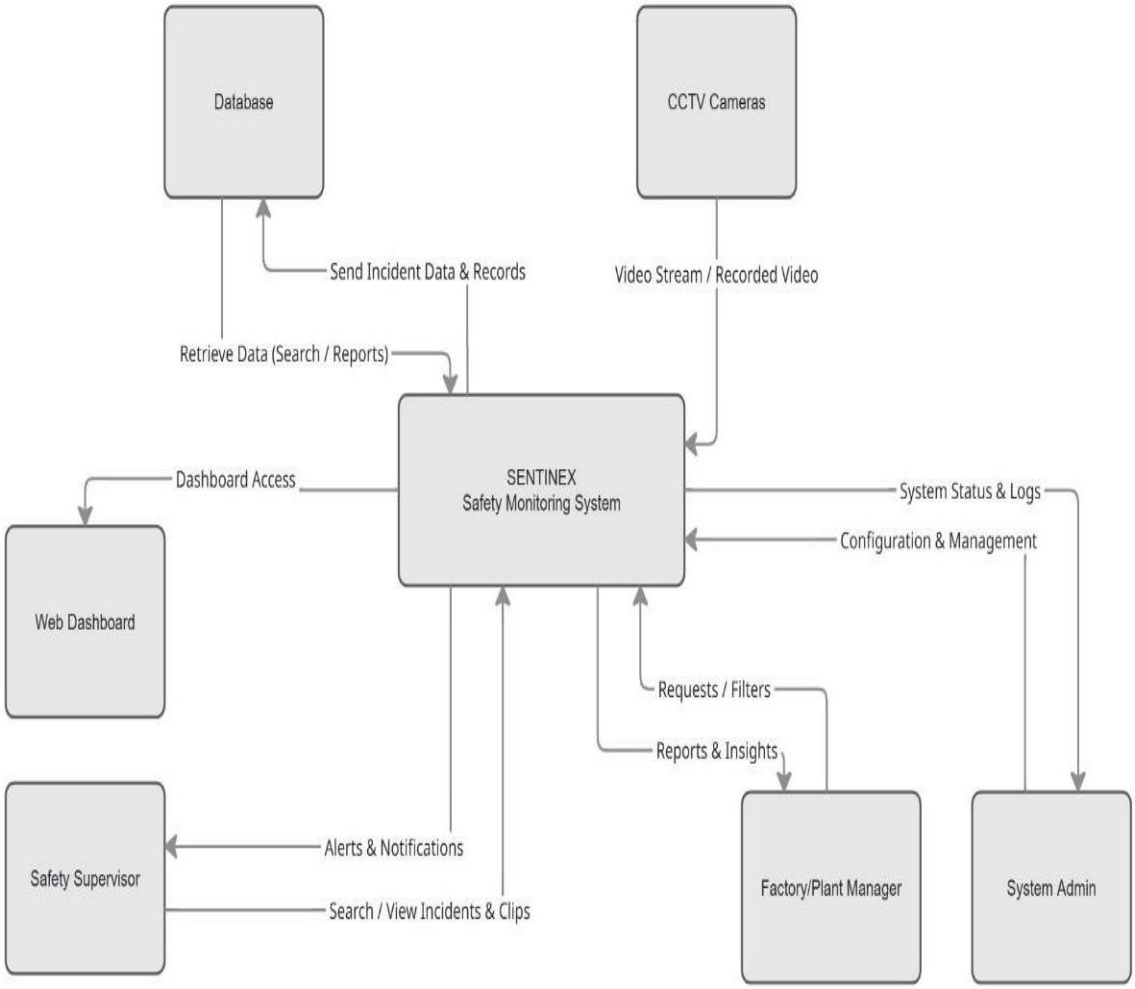


Figure 13: Architecture Diagram

7.2. Package Diagram

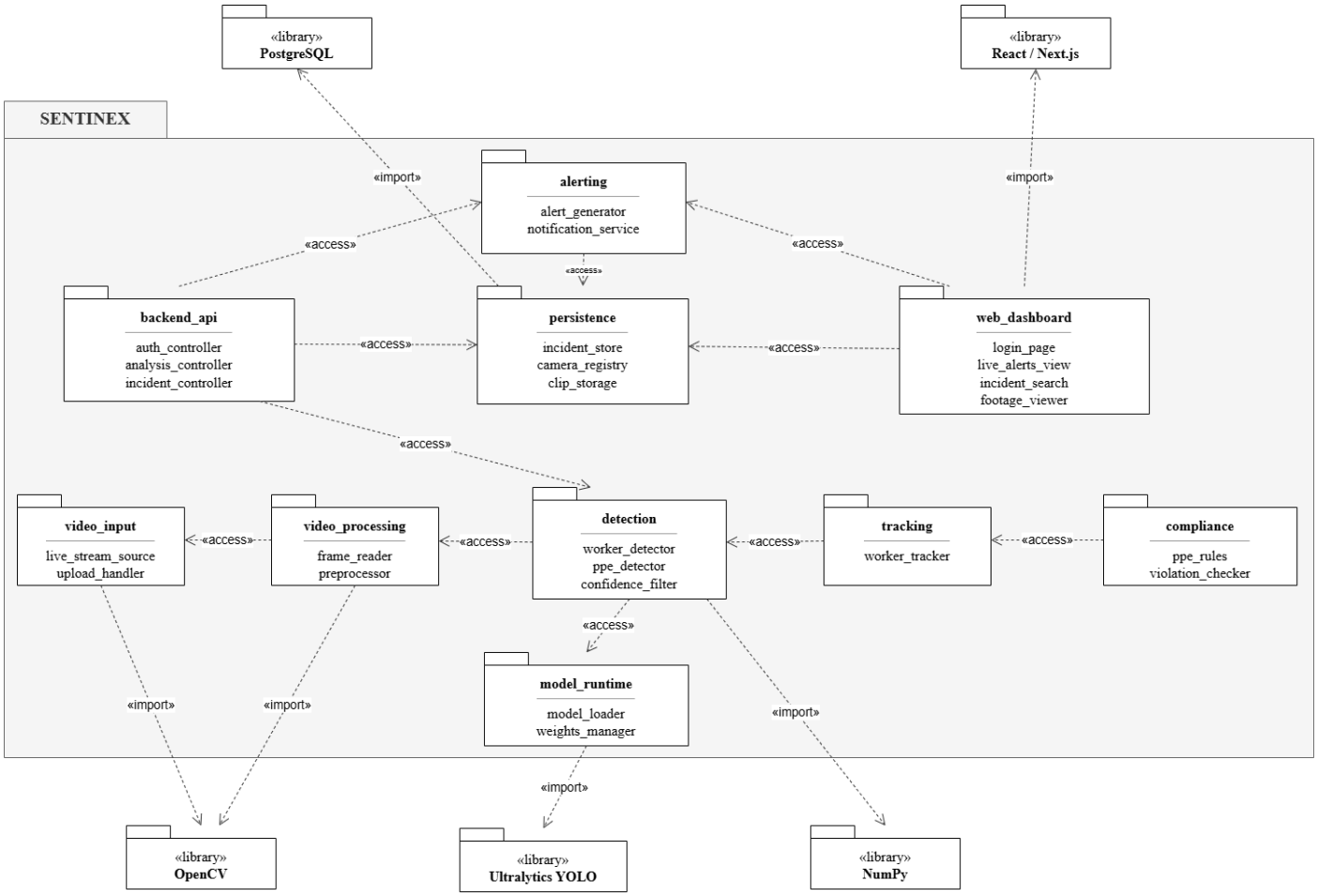


Figure 14: Package Diagram

7.3. Domain Model

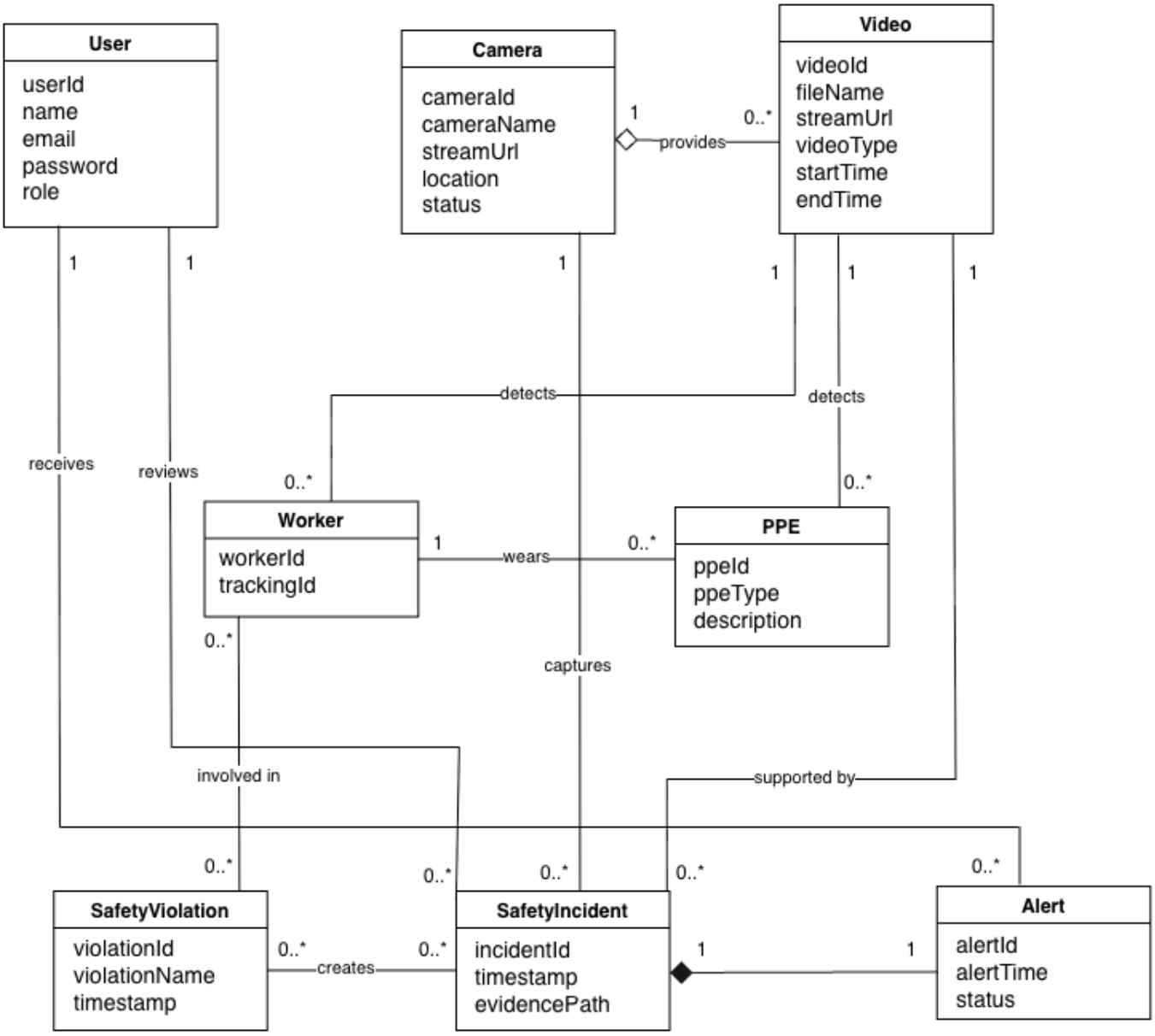


Figure 15: Domain Model

7.4. Entity Relationship Diagram (ERD)

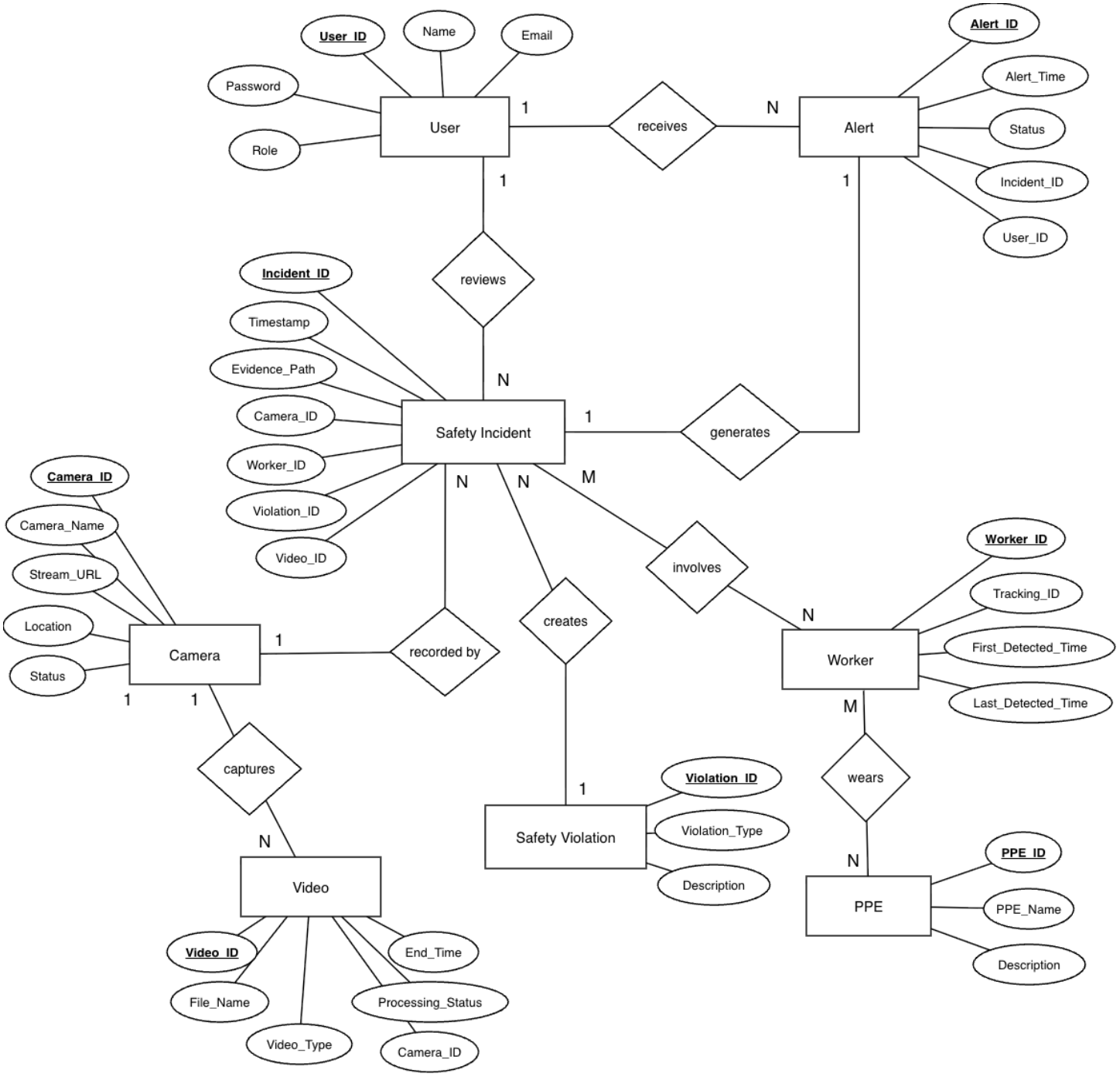


Figure 16: Entity Relationship Diagram

7.5. Component Diagram

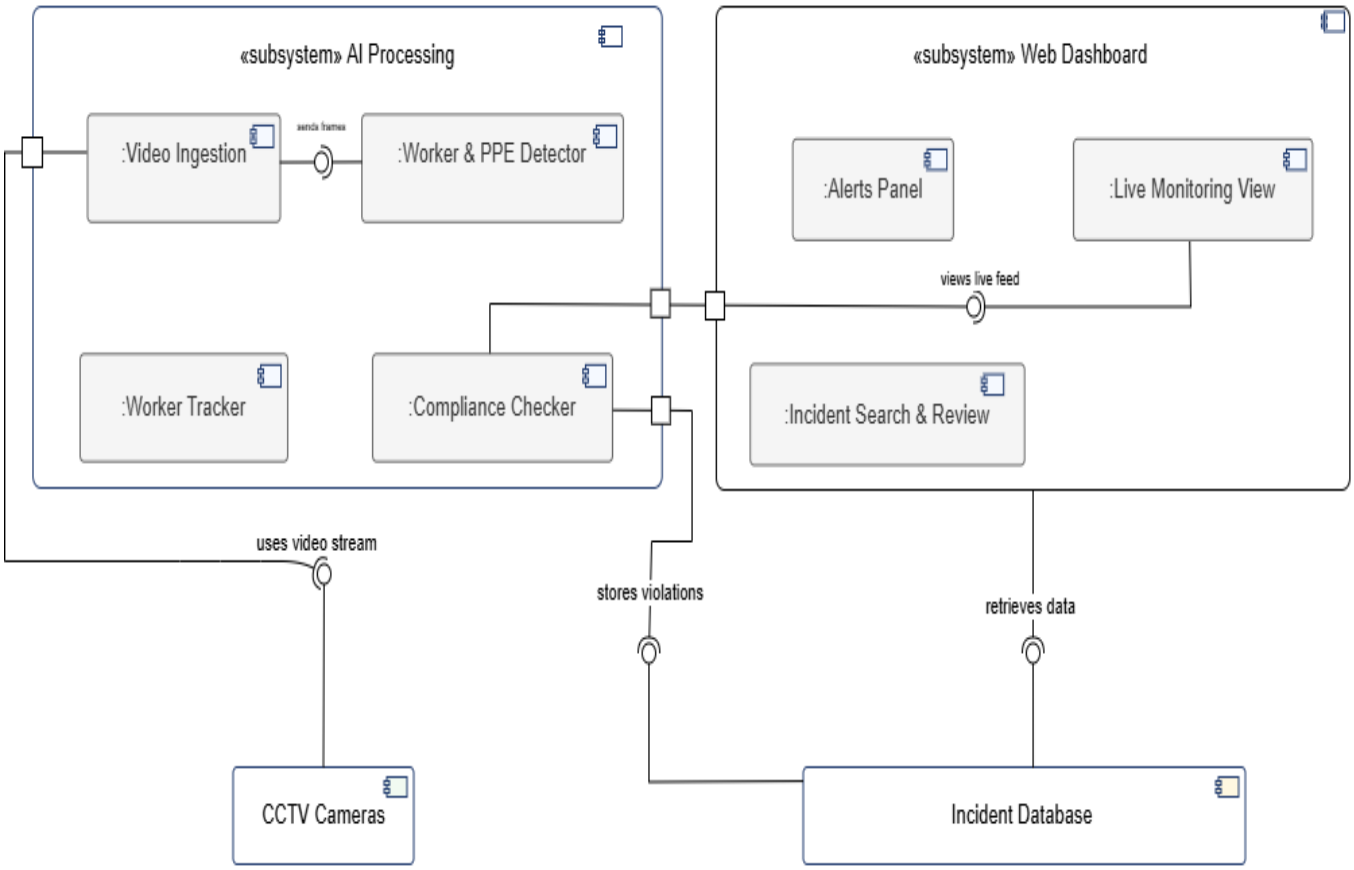


Figure 17: Component Diagram

7.6. Deployment Diagram

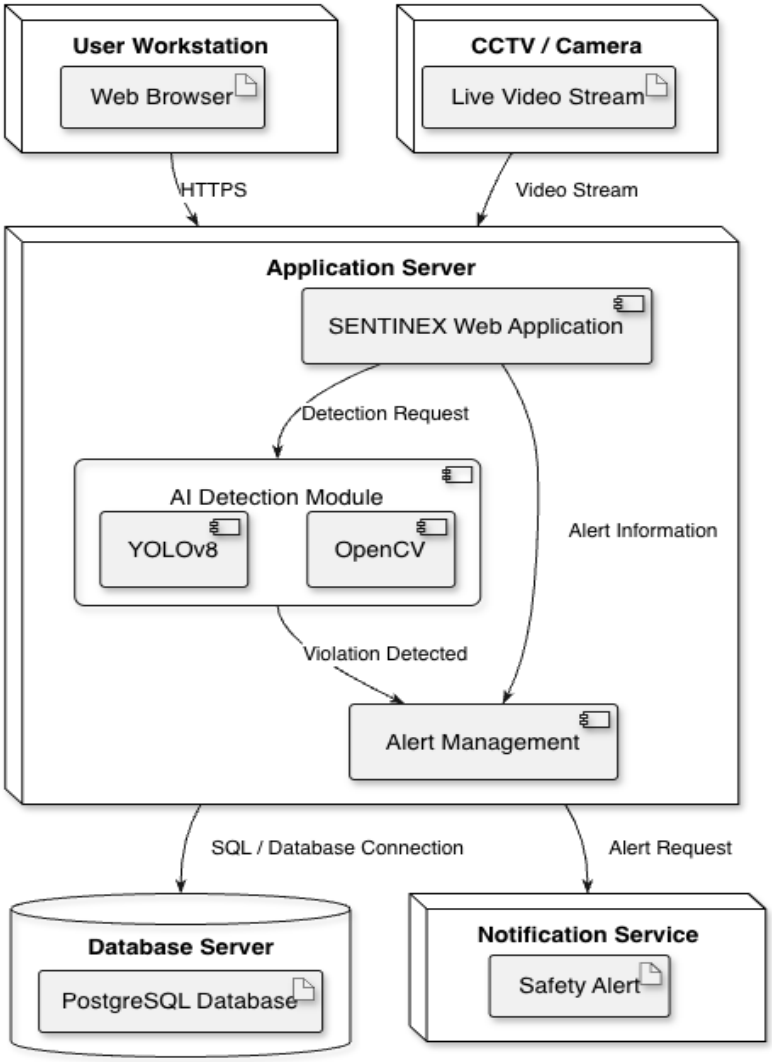


Figure 18: Deployment Diagram

8. Behavioral Design

8.1. Data Flow Diagram

8.1.1. Context Diagram

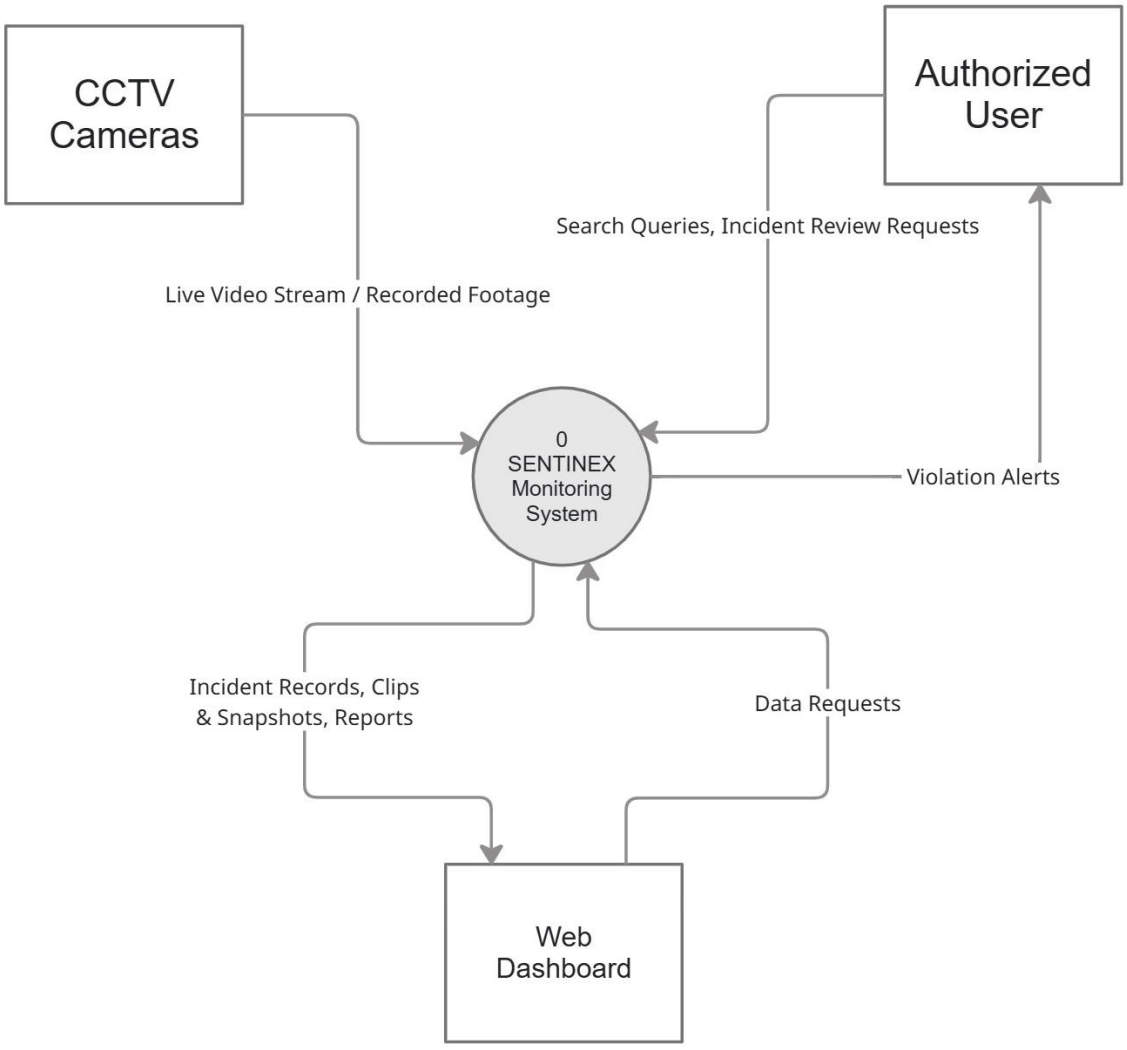


Figure 19: DFD LEVEL 0

8.1.2. Level 1

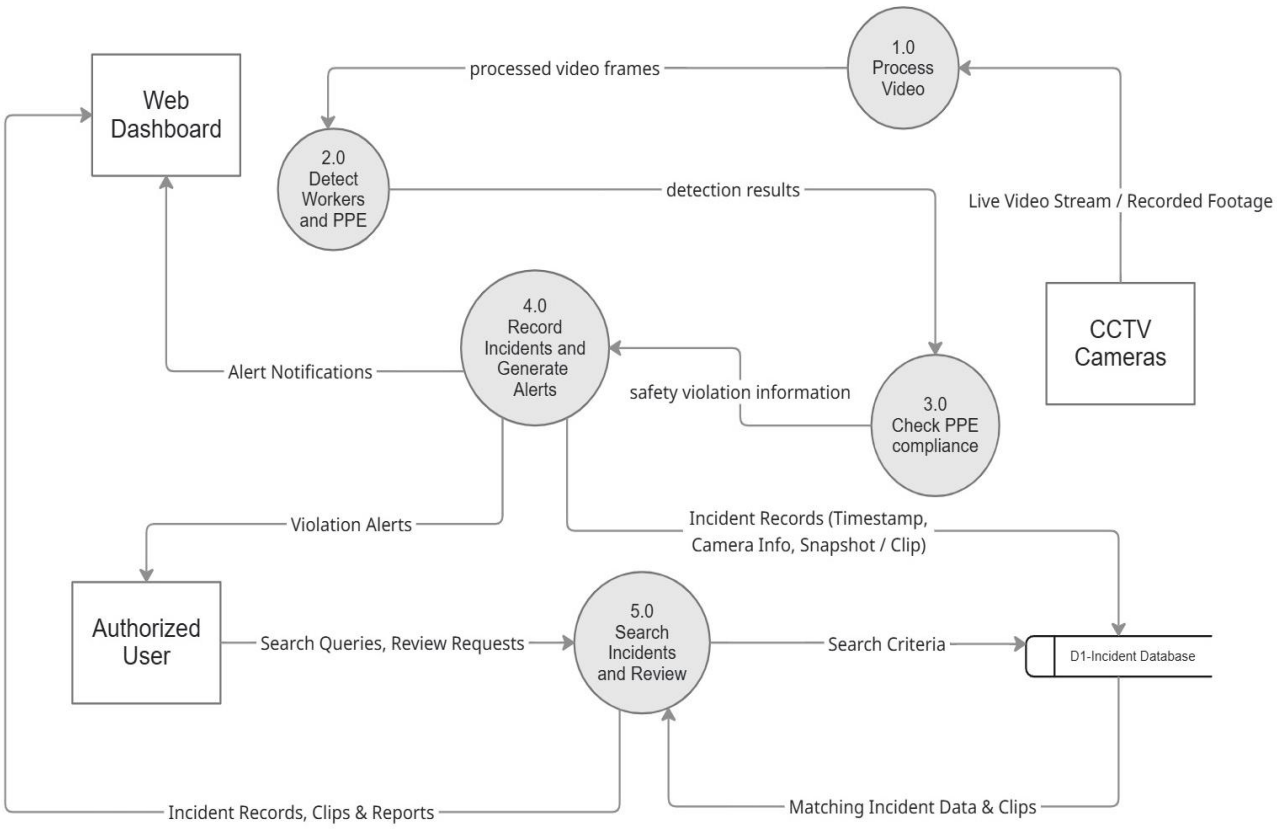


Figure 20: DFD LEVEL 1

8.1.3. Level 2

8.1.3.1. Processes 1, 2, 3: Video Processing and Compliance Checking

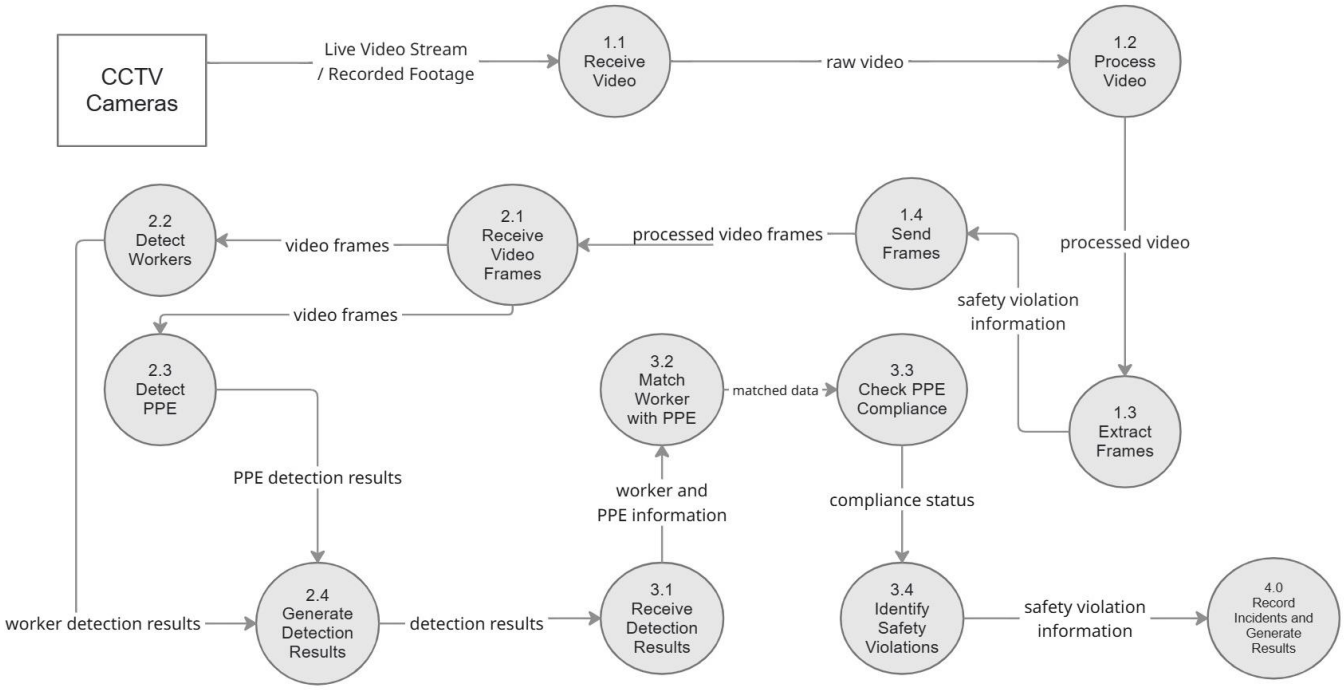


Figure 21: DFD LEVEL 2 – Process 1,2,3

8.1.3.2. Process 4: Incident Recording and Alert Generation

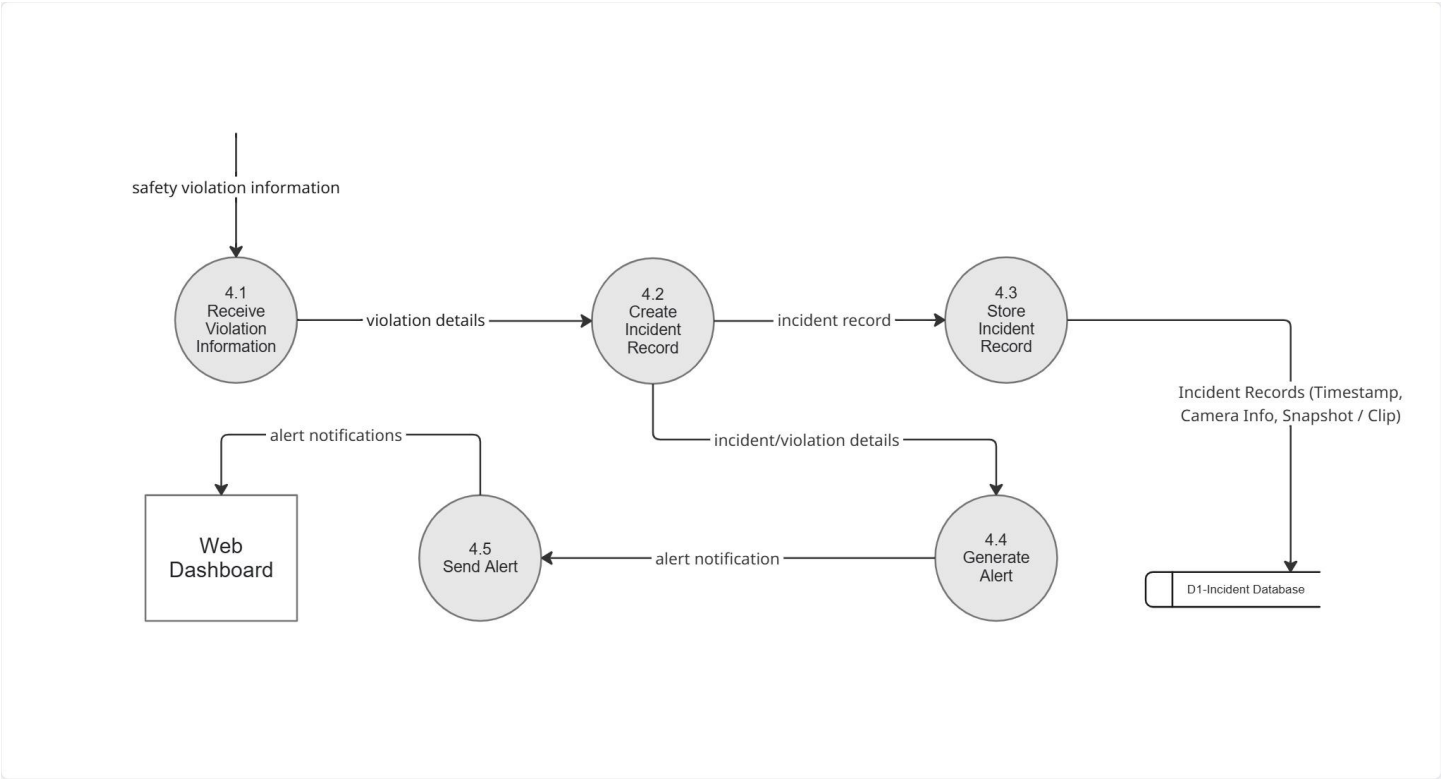


Figure 22: DFD LEVEL 2 – Process 4

8.1.3.3. Process 5: Incident Search and Review

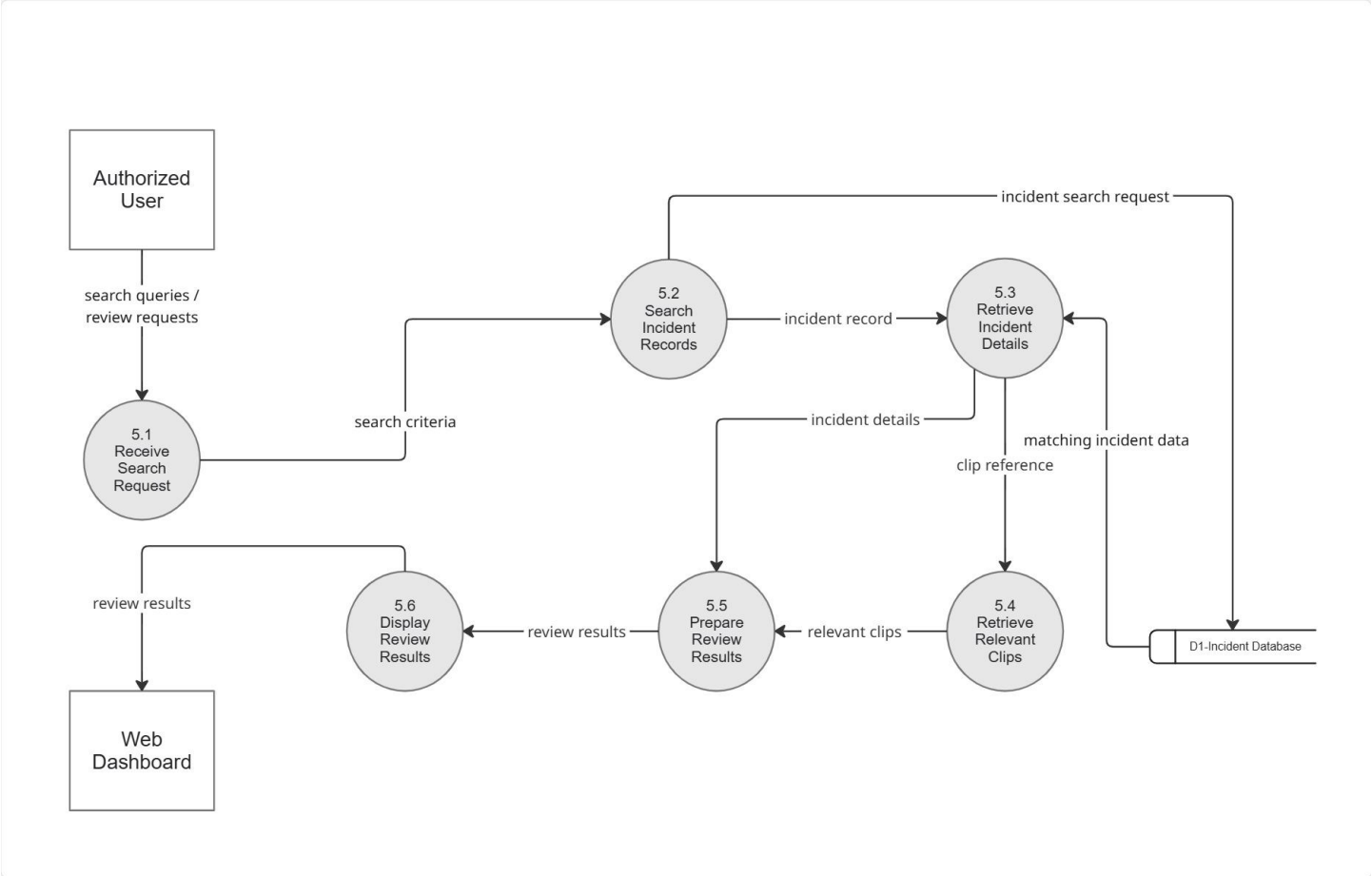


Figure 23: DFD LEVEL 2 – Process 5